Traceability Matrix

REMEMBER GORDON! – Rulebook ⇌ Implementation Mapping

Generated from `docs/traceability.toml`

Overview

Implemented	Descriptive	Implicit	Out-of-scope
84	33	4	3

Total mappings: 124 · Total impl sites: 251

Table of Contents

0/3 implemented

§1 – Introduction

§1 – Introduction

descriptive

manual page unknown

> Introduction

§1.1 – General Comments

descriptive

manual page 1

> General Comments
>
> "REMEMBER GORDON!" — THE BATTLE OF OMDURMAN is a simulation of the final battle in Great Britain's two-year campaign to reassert her presence in the Sudan (1896–1898). Fought September 2nd, 1898, Omdurman finally broke the back of the fanatical Dervish rebellion and gained Britain a million square miles of desolate territory and two million impoverished subjects. With two players, one assumes the role of Herbert Kitchener, Sirdar (CIC) of the Anglo-Egyptian army; the other player becomes the Khalifa, Abdullah the Taiasha, absolute ruler of the Dervishes. The game is also suited for multi-player participation, with each player assuming command of one or more Dervish tribes or Anglo-Egyptian brigades.
>
> While "REMEMBER GORDON!" — THE BATTLE OF OMDURMAN is not, strictly speaking, a beginner's game, the mechanics of play should be familiar to players of modest experience. It is suggested that the bonus game, FALL OF KHARTOUM, and the historical scenario be played first to familiarize players with the game system prior

(see manual for full text)

§1.2 – Game Scale

descriptive

manual page unknown

> Game Scale
>
> Each hexagon of the mapsheet represents approximately 400–440 yards of real terrain and each day turn is the equivalent of two hours of real time. Each counter of infantry and cavalry represents between 400 and 700 men, and each of the gunboats present at the battle has its own counter. The upper echelon of command is represented by individual leader counters for the Anglo-Egyptian force; and leaders plus their retinues for the Dervish army.

`3/7 implemented`

§2 – Game Components

§2 – Game Components

descriptive

manual page unknown

> Game Components

§2.1 – The Game Maps

descriptive

manual page unknown

> The Game Maps
>
> The Omdurman battle map represents approximately 100 square miles of real territory and includes the area north of Omdurman in which the historical battle took place as well as the dominant terrain features that influenced the course of the battle. Note that the mapsheet also contains the Turn Record Track, Turn Sequence, and Terrain Effects Chart at the top; and the Combat Tables and Howitzer Fire Scattergram in the lower right corner. The large letters "A", "D", "Y", etc. are set-up hexes for the historical scenario only (9.2) and should be ignored in the campaign game. Similarly, the hexsides of the Zariba exist only in the historical scenario and should be considered clear terrain in the campaign game. Note, however, that the Anglo-Egyptian player may "construct" the Zariba in the campaign game if desired (see 5.3). All full hexes of the Omdurman game map are playable, including the seven hexes of the Howitzer Fire Scattergram.
>
> The mini-map for the bonus game, FALL OF KHARTOUM, represents that city as it appeared in 1885. The portion

(see manual for full text)

See also: §5.3, §9.2

§2.2 – Play Aids

descriptive

manual page unknown

> Play Aids
>
> Certain charts and tables are needed to play the game. The Terrain Effects Chart lists all terrain found on the mapsheet and the effect of each type on movement and combat. The Combat Tables describe the range effects on various weapon types and includes the Combat Results Table. Also note the Line of Sight Table on the back of this rulebook. It tells players when certain terrain types block line of sight, thus preventing direct fire attacks on enemy units. Players should become familiar with these various charts and tables prior to the beginning of play.

(see manual for full text)

§2.3 – The Units

implemented

manual page unknown

> The Units

File	Symbol	Code Snippet

omdurman-types/src/lib.rs:871 GH:omdurman-types/src/lib.rs:871	**UnitKind**	`869 /// `Some(UnitKind::Marker)` or `None`.` `870 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)]` `871 pub enum UnitKind {` `872     /// Foot infantry (§2.3): fire / melee / movement.` `873     Infantry {`
omdurman-rules/src/lib.rs:798 GH:omdurman-rules/src/lib.rs:798	**UnitProfile**	`796 /// print no melee value).` `797 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)]` `798 pub struct UnitProfile {` `799     pub kind: UnitKind,` `800     pub identity: UnitIdentity,`
omdurman-rules/src/lib.rs:16 GH:omdurman-rules/src/lib.rs:16	**BrigadeId**	`14` `15 use omdurman_types::{` `16     BrigadeId, BrigadeNationality, DayNight, DervishTribe, Faction, HexCoord, HexsideRef, Player,` `17     SetupLetter, UnitKind,` `18 };`
omdurman-types/src/lib.rs:679 GH:omdurman-types/src/lib.rs:679	**SpriteAnnotation**	`677 /// as an optional overlay over the compiled `sprite_data` fallback.` `678 #[derive(Serialize, Deserialize, Clone, Debug, PartialEq)]` `679 pub struct SpriteAnnotation {` `680     pub color: SpriteColor,` `681     pub faction: Option<Faction>,`

Covered by tests: **omdurman-rules::src::unit_profiles::british_army_row_zero_specials_classify_by_counter** **omdurman-rules::src::unit_profiles::egyptian_army_row_zero_specials_classify_by_counter** **omdurman-rules::src::unit_profiles::tribe_stats_come_from_annotation** **omdurman-rules::src::unit_profiles::section_owner_dervish_sections** **omdurman-rules::src::unit_profiles::section_owner_green_sections_are_dervish**

§2.4 – Game Parts Inventory

out-of-scope

manual page unknown

> Game Parts Inventory
>
> Your complete copy of "REMEMBER GORDON!" — THE BATTLE OF OMDURMAN includes:
>
> - One 22 × 28 Battle of Omdurman mapsheet
> - One 8½ × 11 bonus game: FALL OF KHARTOUM mapsheet
> - One Rules Booklet
> - One die-cut Unit Counter Sheet
> - One Campaign Game Order of Appearance Card
> - One ten-sided die
> - One game box

§2.31 – Dervish weapon types

implemented

manual page unknown

> Dervish artillery, gunboats, and forts fire on the "artillery" line of the Dervish Range Effects Table; Jehadia and Danagla units fire on the "rifles" line as does the Isa Zachneih unit. All other Dervish units (including leaders) are armed with spears and swords.

File	Symbol	Code Snippet

omdurman-rules/src/lib.rs:476 GH:omdurman-rules/src/lib.rs:476	**WeaponClass**	`474 |     Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Hash, Debug, strum::Display,` `475 | )]` `476 | pub enum WeaponClass {` `477 |     /// Dervish spears and swords -- no ranged fire at all.` `478 |     Melee,`
omdurman-rules/src/unit_profiles.rs:518 GH:omdurman-rules/src/unit_profiles.rs:518	**dervish_tribe**	`516 | /// Resolve a Dervish tribal foot counter (§2.31): Jehadia, Danagla and` `517 | /// Isa Zachneih fire on the rifles line; every other tribe is spear-armed.` `518 | pub fn dervish_tribe(tribe: DervishTribe) -> Option<Classification> {` `519 |     // §2.31: "Jehadia and Danagla units fire on the 'rifles' line as does the` `520 |     // Isa Zachneih unit. All other Dervish units (including leaders) are armed`
omdurman-rules/src/unit_profiles.rs:292 GH:omdurman-rules/src/unit_profiles.rs:292	**khalifa_abdullah**	`290 | ///     battle (§9.322). All three are interchangeable, so they share the` `291 | ///     `DervishArtillery` identity.` `292 | pub fn khalifa_abdullah(col: u32, row: u32) -> Option<Classification> {` `293 |     let artillery = || {` `294 |         Some(Classification {`

Covered by tests: **omdurman-rules::src::unit_profiles::dervish_weapon_class_follows_the_rifles_line**

§2.32 – Anglo-Egyptian weapon types

implemented

manual page unknown

> All Anglo-Egyptian units (except gunboats, Maxims, artillery, and leaders) are armed with rifles. Maxims fire on the "Maxims" line of the Anglo-Egyptian Range Effects Table, and artillery and old gunboats fire on the "Artillery" line. New type (named) gunboats may fire on the "Howitzer", "Artillery", and "Maxims" lines of the Range Effects Table. (See 6.52 for the fire capabilities of the "Friendlies".)
>
> **Sample Dervish Units** (printed on counters): combat unit (Combat / Melee / Movement values, plus Tribe identifier); Leader (e.g. OSMAN DIGNA); Camel unit (e.g. Danagla, 4-6-12); Fort.
>
> **Sample Anglo-Egyptian Units** (printed on counters): Cavalry (e.g. 21 Lancers); Artillery (e.g. 32 Battery); Old Gunboat (e.g. LORD KITCHENER, 0-0-15); New Gunboat — named (e.g. Sultan, with artillery and howitzer factor, plus movement downstream / movement upstream values); Maxim Guns (fire twice per turn); Infantry (Fire Combat Factor / Melee / Movement, plus Battalion ID and Brigade ID — e.g. "2B" = 2nd British Brigade, "3E" = 3rd Egyptian Brigade).

(see manual for full text)

See also: §6.52

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:428 GH:omdurman-rules/src/lib.rs:428	**GunboatId**	`426 | /// fire; "old" gunboats do not (rulebook §2.32).` `427 | #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)]` `428 | pub enum GunboatId {` `429 |     /// One of the five new-type named gunboats with howitzer capability.` `430 |     Named(NamedGunboat),`
omdurman-rules/src/lib.rs:447 GH:omdurman-rules/src/lib.rs:447	**NamedGunboat**	`445 | /// The five named gunboats with howitzer capability (rulebook §6.64, §2.32).` `446 | #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)]` `447 | pub enum NamedGunboat {` `448 |     Sultan,` `449 |     Melik,`

omdurman-rules/src/lib.rs:460 GH:omdurman-rules/src/lib.rs:460	**OldGunboat**	`458 │ /// in the Maxim Second Fire and Howitzer subphase (§6.42).` `459 │ #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)]` `460 │ pub enum OldGunboat {` `461 │     LordKitchener,` `462 │     Tamai,`
omdurman-rules/src/lib.rs:460 GH:omdurman-rules/src/lib.rs:460	**GunboatId::Old**	`458 │ /// in the Maxim Second Fire and Howitzer subphase (§6.42).` `459 │ #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)]` `460 │ pub enum OldGunboat {` `461 │     LordKitchener,` `462 │     Tamai,`
omdurman-rules/src/lib.rs:434 GH:omdurman-rules/src/lib.rs:434	**GunboatId::DervishGunboat**	`432 │     Old(OldGunboat),` `433 │     /// A Dervish gunboat (§9.111, §10.14).` `434 │     DervishGunboat(u8),` `435 │ }` `436 │`

Covered by tests: **omdurman-rules::src::effects::tests::old_gunboat_lacks_howitzer** **omdurman-rules::src::effects::tests::old_gunboat_rejected_from_howitzer_subphase**

`0/1 implemented`

§3 – Getting Started

§3 – Getting Started

out-of-scope

manual page unknown

> Getting Started
>
> Spread out the mapsheet on a table. It should lie flat if you backfold it against the scored lines. The Dervish player should sit next to the west edge of the map and the Anglo-Egyptian player opposite him on the east edge. Read through the rules once, looking over the various charts as they are referred to in the various sections. Next, select a scenario and punch out only those unit counters needed to play. Later on, the rest of the unit counters should be punched out, sorted and stored by unit type.

(see manual for full text)

`1/1 implemented`

§4 – Turn Sequence

§4 – Turn Sequence

implemented

manual page unknown

> Turn Sequence
>
> “REMEMBER GORDON!” — THE BATTLE OF OMDURMAN is played in “Game Turns”, each of which has two “Player Turns”. The player moving first will vary according to the scenario being played. In the Campaign Game, for example, the Anglo-Egyptian player moves first.
>
> ### A) Anglo-Egyptian Player Turn:
>
> 1. Anglo-Egyptian Movement Phase
> 2. Fire Combat Phase a. Dervish Defensive Fire b. Anglo-Egyptian Offensive Fire
>   1. Direct Fire Subphase
>   2. Maxim Second Fire and Howitzer Fire Subphase

(see manual for full text)

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:252 GH:omdurman-rules/src/lib.rs:252	**GameTurnIndex**	`250 │ /// One-based Game Turn index (1, 2, ... up to the scenario length) (rulebook §4).` `251 │ #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Hash, Debug)]` `252 │ pub struct GameTurnIndex(u8);` `253 │` `254 │ impl GameTurnIndex {`
omdurman-rules/src/lib.rs:274 GH:omdurman-rules/src/lib.rs:274	**Phase**	`272 │ /// etc.` `273 │ #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug, Default)]` `274 │ pub enum Phase {` `275 │     /// Pre-game deployment (§9.2/§9.3/§10): fixed units are placed, each side` `276 │     /// deploys its order of battle within its legal zone, and river`
omdurman-rules/src/effects/state.rs:5 GH:omdurman-rules/src/effects/state.rs:5	**GameState**	`3 │ /// All mutable state of a game in progress (rulebook §4).` `4 │ #[derive(Serialize, Deserialize, Clone, Debug)]` `5 │ pub struct GameState {` `6 │     pub scenario: Scenario,` `7 │     pub current_turn: GameTurnIndex,`
omdurman-rules/src/effects/state.rs:141 GH:omdurman-rules/src/effects/state.rs:141	**GameState::new**	`139 │ impl GameState {` `140 │     /// Create a fresh game state for a given scenario (rulebook §4).` `141 │     pub fn new(scenario: Scenario) -> Self {` `142 │         let first = scenario_turn(scenario, GameTurnIndex::new(1));` `143 │         // First player to *move* per scenario: Campaign -- Anglo-Egyptian moves`
omdurman-rules/src/effects/effect.rs:17 GH:omdurman-rules/src/effects/effect.rs:17	**AdvancePhase**	`15 │     /// At end-of-turn, disrupted units recover and per-turn tracking is` `16 │     /// cleared.` `17 │     AdvancePhase,` `18 │` `19 │     // -- Movement ---------------------------------------------------------`
omdurman-rules/src/effects/dispatch.rs:98 GH:omdurman-	**advance_phase**	`96 │` `97 │ /// Advance the game state to the next phase (rulebook §4).`

rules/src/effects/ dispatch.rs:98		`98 │ pub fn advance_phase(state: &mut GameState) -> Result<(), RuleError> {` `99 │     let old_phase = state.phase;` `100 │     #[cfg(not(feature = "kani"))]`
omdurman-rules/src/effects/dispatch.rs:204 GH:omdurman-rules/src/effects/dispatch.rs:204	**end_player_turn**	`202 │` `203 │ /// End the current player's turn: recover disrupted units, switch active player, advance turn index (rulebook §4).` `204 │ pub fn end_player_turn(state: &mut GameState) -> Result<(), RuleError> {` `205 │     #[cfg(not(feature = "kani"))]` `206 │     debug!(`
omdurman-rules/src/lib.rs:63 GH:omdurman-rules/src/lib.rs:63	**GameTurnIndex::value**	`61 │             pub const ALL: &'static [Self] = &[$(Self::$variant,)+];` `62 │` `63 │             pub fn value(self) -> u16 {` `64 │                 match self {` `65 │                     $(Self::$variant => $value,)+`
omdurman-rules/src/effects/state.rs:133 GH:omdurman-rules/src/effects/state.rs:133	**PendingMelee**	`131 │ /// resolution after the reaction window is deterministic and host-ordered (rulebook §7.5).` `132 │ #[derive(Serialize, Deserialize, Clone, Debug)]` `133 │ pub struct PendingMelee {` `134 │     pub attack: MeleeAttack,` `135 │     pub attacker_roll: DieRoll,`

Covered by tests: **omdurman-rules::src::effects::tests::both_ready_auto_advances_out_of_setup** **omdurman-rules::src::effects::tests::fire_combat_wrong_phase_rejected** **omdurman-rules::src::effects::tests::new_game_starts_in_setup** **omdurman-rules::src::turn_track::scenario_turn_dispatches_correctly** **omdurman-rules::src::effects::tests::turn_advances_through_phases** **omdurman-rules::src::unit_profiles::game_turn_marker_cell_returns_none**

18/23 implemented

§5 – Movement Phase

§5 – Movement Phase (general)

descriptive

manual page unknown

> Movement Phase

§5.1 – General Rules

descriptive

manual page unknown

> General Rules

§5.2 – Movement Restrictions

descriptive

manual page unknown

> Movement Restrictions

§5.3 – Constructing the Zariba

implemented

manual page unknown

> Constructing the Zariba
>
> The Zariba trench and thorn hedge hexsides are built and in place in the historical scenario only. These hexsides are considered clear terrain in the campaign game. The Anglo-Egyptian player may, however, find it useful to construct this defensive position during the campaign game. The Zariba hexsides may only be built in their position as displayed on the mapsheet. Construction procedure is as follows: any Anglo-Egyptian infantry unit that begins and ends the Anglo-Egyptian player turn adjacent to (and on the Nile side of) Zariba hexsides has constructed all Zariba hexsides to which he is adjacent. The constructing unit may neither fire offensively nor melee attack during the turn of construction. Use a blank counter to denote units constructing Zariba hexsides. See 9.23 for defensive benefits and movement restrictions of Zariba hexsides.

(see manual for full text)

See also: §9.23

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:832 GH:omdurman-rules/src/lib.rs:832	**constructing_zariba**	`830 │     /// Set while the unit is building Zariba hexsides -- neither offensive` `831 │     /// fire nor melee allowed that turn (§5.3).` `832 │     pub constructing_zariba: bool,` `833 │     /// Set when the Royal Engineers are committed to a demolition this turn` `834 │     /// (§6.53) -- neither offensive fire nor melee allowed that turn.`
omdurman-rules/src/effects/effect.rs:169 GH:omdurman-	**ConstructZariba**	`167 │` `168 │     /// Begin constructing a Zariba hexside (rulebook §5.3).` `169 │     ConstructZariba {`

rules/src/effects/effect.rs:169		`170 │         unit_ids: Vec<UnitId>,` `171 │         hexside: HexsideRef,`
omdurman-rules/src/effects/dispatch.rs:40 GH:omdurman-rules/src/effects/dispatch.rs:40	apply_construct_zariba	` 38 │         GameEffect::RecoverUnit { unit_id } => apply_recover_unit(state, *unit_id),` ` 39 │         GameEffect::ConstructZariba { unit_ids, hexside } => {` ` 40 │             apply_construct_zariba(state, unit_ids, *hexside)` ` 41 │         }` ` 42 │         GameEffect::Demolition { unit_id, target } => apply_demolition(state, *unit_id, *target),`

Covered by tests: **omdurman-rules::src::effects::tests::construct_zariba_marks_builders_and_records_hexside**

§5.4 – Zones of Control

descriptive

manual page unknown

> Zones of Control

§5.5 – Stacking

descriptive

manual page unknown

> Stacking

§5.11 – Movement allowances printed on units

implemented

manual page unknown

> The movement allowances of the various unit types are printed directly on the units (see 2.3). A unit may move up to this printed movement allowance, paying varying costs for different terrain types (see the Terrain Effects Chart).

See also: §2.3

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:131 GH:omdurman-rules/src/lib.rs:131	MovementAllowance	`129 │     /// is a named variant.` `130 │     #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug)]` `131 │     pub enum MovementAllowance {` `132 │         /// Immobile (forts, wrecked gunboats).` `133 │         Immobile = 0,`
omdurman-rules/src/lib.rs:809 GH:omdurman-rules/src/lib.rs:809	UnitMovement	`807 │ /// Movement allowance -- uniform for land units, split for gunboats (rulebook §5.11, §5.24, §5.25).` `808 │ #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)]` `809 │ pub enum UnitMovement {` `810 │     Land(MovementAllowance),` `811 │     Gunboat(GunboatMovement),`
omdurman-types/src/lib.rs:302 GH:omdurman-types/src/lib.rs:302	HexDirection	``300 │ /// (`+q`, `+q+r`, `+r`, `-q`, `-q-r`, `-r` for pointy-top hexes) (rulebook §5.11, §5.24).`` `301 │ #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, Default)]` `302 │ pub enum HexDirection {` `303 │     #[default]` `304 │     East = 0,`

omdurman-rules/src/lib.rs:173 GH:omdurman-rules/src/lib.rs:173	**MovementPoints**	Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Hash, Debug, Default,)] pub struct MovementPoints(i16); impl MovementPoints {
omdurman-rules/src/terrain_chart.rs:21 GH:omdurman-rules/src/terrain_chart.rs:21	**terrain_effects_chart**	/// /// Source: printed Terrain Effects Chart on the mapsheet. pub fn terrain_effects_chart(terrain: Terrain) -> TerrainEntry { match terrain { Terrain::Clear { .. } => TerrainEntry {
omdurman-rules/src/terrain_chart.rs:77 GH:omdurman-rules/src/terrain_chart.rs:77	**movement_cost**	/// Convenience: get the movement cost for a terrain type (rulebook §5.11, Terrain Effects Chart). /// Returns `None` for impassable terrain (Nile). pub fn movement_cost(terrain: Terrain) -> Option<MovementAllowance> { terrain_effects_chart(terrain).movement_cost }
omdurman-rules/src/terrain_chart.rs:85 GH:omdurman-rules/src/terrain_chart.rs:85	**movement_cost_with_ro**	/// underlying terrain; without a road it's the terrain's own cost. The road is /// a movement overlay only -- combat/LOS still use the underlying terrain. pub fn movement_cost_with_road(terrain: Terrain, road: bool) -> Option<MovementAllowance> { if road { Some(MovementAllowance::One)
omdurman-types/src/lib.rs:498 GH:omdurman-types/src/lib.rs:498	**Terrain::has_road**	 /// Whether this hex has any road touching it. pub fn has_road(self) -> bool { !matches!(self.road(), Road::None) }
omdurman-types/src/lib.rs:435 GH:omdurman-types/src/lib.rs:435	**Terrain::passable_by_lan**	 /// Whether this terrain may be entered by land units (rulebook §5.11). pub fn passable_by_land(self) -> bool { !self.is_nile() }
omdurman-types/src/lib.rs:503 GH:omdurman-types/src/lib.rs:503	**Terrain::is_crossroad**	 /// Whether roads converge at this hex's centre. pub fn is_crossroad(self) -> bool { matches!(self.road(), Road::Crossroad) }
omdurman-hexmap/src/map.rs:17 GH:omdurman-hexmap/src/map.rs:17	**GameMap::roads**	pub hexes: HashMap<HexCoord, HexData>, pub hexsides: HashMap<HexsideRef, HexsideKind>, pub roads: HashSet<HexsideRef>, pub excluded: HashSet<HexCoord>, pub overlay: OverlayParams,

Covered by tests: **omdurman-rules::src::terrain_chart::clear_terrain_no_bonus** **omdurman-rules::src::terrain_chart::nile_is_impassable** **omdurman-rules::src::terrain_chart::rough_movement_and_defense** **omdurman-rules::src::terrain_chart::swamp_movement_and_defense** **omdurman-rules::src::terrain_chart::hilltop_movement_and_defense** **omdurman-rules::src::terrain_chart::huts_movement_and_defense** **omdurman-rules::src::terrain_chart::movement_cost_convenience_matches_chart** **omdurman-rules::src::terrain_chart::movement_cost_with_road_always_one** **omdurman-rules::src::effects::tests::land_unit_may_not_enter_nile** **omdurman-rules::src::terrain_chart::movement_cost_without_road_matches_terrain** **omdurman-rules::src::effects::tests::movement_cost_for_uses_terrain** **omdurman-rules::src::effects::tests::movement_cost_for_road_costs_one** omdurman-rules::src::terrain_chart::road_gives_crossroad omdurman-rules::src::terrain_chart::terrain_movement_costs_in_bounds omdurman-rules::src::terrain_chart::terrain_chart_road_always_costs_one

§5.12 – Move up to allowance, hex by hex (cumulative MP cap)

implemented

manual page unknown

> A player may move as many or as few of his units as desired during each movement phase, limited only by the units' movement allowance, the terrain costs paid in moving from hex to hex, and enemy zones of control (see 5.4).

See also: §5.4

File	Symbol	Code Snippet
omdurman-rules/src/effects/state.rs:665 GH:omdurman-rules/src/effects/state.rs:665	**can_move_unit**	`` 663 │     /// the same `RuleError` the `MoveUnit` effect would on rejection. Lets the `` `664 │     /// UI gate input without mutating or duplicating the rules.` `665 │     pub fn can_move_unit(&self, unit_id: UnitId, cost: MovementPoints) -> Result<(), RuleError> {` `666 │         self.can_move_unit_to(unit_id, None, cost)` `667 │     }`
omdurman-rules/src/effects/state.rs:1356 GH:omdurman-rules/src/effects/state.rs:1356	**mp_spent**	`1354 │` `` 1355 │     /// Movement points `unit_id` has already spent this turn (§5.11/§5.12). `` `1356 │     pub fn mp_spent(&self, unit_id: UnitId) -> i16 {` `1357 │ self.mp_spent_this_turn.get(&unit_id).copied().unwrap_or(0)` `1358 │     }`

Covered by tests: omdurman-rules::src::effects::tests::cumulative_move_cost_may_not_exceed_allowance

§5.13 – No MP accumulation between turns

implemented

manual page unknown

> A unit may never accumulate movement points from turn to turn, nor may a unit transfer unused movement points to other units. A unit's unused movement points in any given turn are considered lost.

File	Symbol	Code Snippet
omdurman-rules/src/effects/dispatch.rs:204 GH:omdurman-rules/src/effects/dispatch.rs:204	**end_player_turn**	`202 │` `203 │ /// End the current player's turn: recover disrupted units, switch active player, advance turn index (rulebook §4).` `204 │ pub fn end_player_turn(state: &mut GameState) -> Result<(), RuleError> {` `205 │     #[cfg(not(feature = "kani"))]` `206 │     debug!(`

Covered by tests: omdurman-rules::src::effects::tests::unused_movement_points_do_not_carry_over

§5.21 – Friendlies transport via gunboat

implemented

manual page unknown

> In general, naval transport missions are not allowed, i.e. gunboats may not carry any land units. The sole exception is that the Anglo-Egyptian player may transport the surviving units of the "Friendlies" brigade from the east bank of the Nile to the west bank after, and only after, the Dervish east bank unit (Isa Zachneih) has been eliminated. The

> transport is accomplished in the following sequence: a) on any turn that a “Friendlies” unit and any Anglo-Egyptian gunboat start their turn adjacent, that unit may load onto (i.e. stack with) the gunboat; b) during the Anglo-Egyptian player’s next turn the gunboat may move to any Nile hex adjacent to a west bank hex (up to the gunboat’s movement allowance); c) on the Anglo-Egyptian player’s third turn the “Friendlies” unit may disembark and move normally, paying the normal terrain cost for the first hex entered. The gunboat may also move normally that turn.

(see manual for full text)

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:629 GH:omdurman-rules/src/lib.rs:629	is_friendlies	627 /// "Friendlies" units obey several special rules (§5.21, §5.23, §6.52, 628 /// §9.14 victory conditions). 629 pub fn is_friendlies(&self) -> bool { 630 matches!(631 self,
omdurman-rules/src/effects/state.rs:2050 GH:omdurman-rules/src/effects/state.rs:2050	friendlies_transport_offe	2048 /// regardless of selection. Pairs with [`GameEffect::FriendliesTransport`] 2049 /// so the UI can offer exactly the action the engine would accept. 2050 pub fn friendlies_transport_offer(&self, selected: Option<UnitId>) -> Option<FriendliesAction> { 2051 match self.friendlies_transport { 2052 None => {
omdurman-rules/src/lib.rs:829 GH:omdurman-rules/src/lib.rs:829	loaded_on	827 pub disrupted: bool, 828 /// `Some(gunboat)` after a "Friendlies" unit loads onto a gunboat (§5.21). 829 pub loaded_on: Option<UnitId>, 830 /// Set while the unit is building Zariba hexsides -- neither offensive 831 /// fire nor melee allowed that turn (§5.3).
omdurman-rules/src/effects/effect.rs:196 GH:omdurman-rules/src/effects/effect.rs:196	FriendliesTransport	194 195 /// Load/disembark the "Friendlies" brigade via gunboat (rulebook §5.21). 196 FriendliesTransport(crate::FriendliesAction), 197 198 // -- Optional rules --
omdurman-rules/src/effects/river.rs:12 GH:omdurman-rules/src/effects/river.rs:12	apply_friendlies_transpo	10 /// unit is freed (a disembarking `MoveUnit` should follow, costed by 11 /// terrain). 12 pub fn apply_friendlies_transport(13 state: &mut GameState, 14 action: FriendliesAction,
omdurman-rules/src/lib.rs:1077 GH:omdurman-rules/src/lib.rs:1077	FriendliesAction	1075 /// tracks each unit–gunboat pair independently. 1076 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 1077 pub enum FriendliesAction { 1078 /// Turn N (the load turn): unit and gunboat started adjacent; unit 1079 /// loads onto (stacks with) the gunboat.
omdurman-rules/src/lib.rs:1097 GH:omdurman-rules/src/lib.rs:1097	TransportState	1095 /// third turn. 1096 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 1097 pub enum TransportState { 1098 /// Turn N (the load turn): unit and gunboat started adjacent; unit 1099 /// loads onto (stacks with) the gunboat.

Covered by tests: **omdurman-rules::src::effects::tests::friendlies_transport_offer_load_requires_prerequisites** **omdurman-rules::src::effects::tests::friendlies_transport_offer_follows_state_machine**

§5.22 – Land units may never enter a Nile River hex

implemented

manual page unknown

> With the exception of 5.21, land units may never enter a Nile River hex. Only gunboats may enter and move along Nile River hexes.

See also: §5.21

File	Symbol	Code Snippet
omdurman-rules/src/effects/state.rs:682 GH:omdurman-rules/src/effects/state.rs:682	**can_move_unit_to**	`680 │     ///` `681 │     /// [`hex_in_enemy_zoc`]: Self::hex_in_enemy_zoc` `682 │     pub fn can_move_unit_to(` `683 │         &self,` `684 │         unit_id: UnitId,`
omdurman-rules/src/effects/state.rs:353 GH:omdurman-rules/src/effects/state.rs:353	**in_deployment_zone**	`351 │     ///   plan / UI rather than this hex predicate. Documented, not silently` `352 │     ///   dropped.` `353 │     pub fn in_deployment_zone(&self, player: Player, hex: HexCoord, is_boat: bool) -> bool {` `354 │         // No board attached -> permissive (unit tests, unbound session).` `355 │         if self.board.terrain.is_empty() {`

Covered by tests: **omdurman-rules::src::effects::tests::campaign_deployment_is_boat_land_exclusive** **omdurman-rules::src::effects::tests::fok_ae_gunboat_deploys_only_on_nile** **omdurman-rules::src::effects::tests::fok_ae_land_unit_rejected_on_nile** **omdurman-rules::src::effects::tests::deploy_via_real_sprite_resolution_matches_engine** **omdurman-rules::src::effects::tests::fok_dervish_land_unit_rejected_on_nile** **omdurman-rules::src::effects::tests::retreat_before_melee_may_not_land_on_nile**

§5.23 – Walled city entry restrictions

implemented

manual page unknown

> Only certain units may enter the walled portion of Omdurman. For the Dervish player these are the Khalifa unit, the three Dervish artillery units, and the Taiasha units (the Khalifa's bodyguard). Any Anglo-Egyptian units that can get to the walled city may enter it (except gunboats and "Friendlies"). Units entering and/or exiting the walled city may only do so through a gate or breach hexside.

File	Symbol	Code Snippet
omdurman-types/src/lib.rs:152 GH:omdurman-types/src/lib.rs:152	**HexsideRef**	`150 │ /// data by [`HexsideRef`].` `151 │ #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug)]` `152 │ pub struct HexsideRef {` `153 │     pub a: HexCoord,` `154 │     pub b: HexCoord,`
omdurman-types/src/lib.rs:187 GH:omdurman-types/src/lib.rs:187	**HexsideKind**	`185 │     strum::EnumIter,` `186 │ )]` `187 │ pub enum HexsideKind {` `188 │     /// City wall (Khartoum, walled city of Omdurman). Blocks LOS, blocks` `189 │     /// movement except across gates/breaches (§5.23), blocks ZOC into the city`
omdurman-types/src/lib.rs:258	**blocks_movement**	`256 │     /// `omdurman-rules`). The trench *end* variants are therefore intentionally` `257 │     /// not blocking.` `258 │     pub fn blocks_movement(self) -> bool {`

GH:omdurman-types/src/lib.rs:258		matches!(self,
omdurman-rules/src/lib.rs:655 GH:omdurman-rules/src/lib.rs:655	**may_enter_walled_city**	/// Taiasha bodyguard may enter. Anglo-Egyptian: any unit that can reach the /// walled city *except* gunboats and "Friendlies". pub fn may_enter_walled_city(&self) -> bool { match self { // §5.23 Dervish: Khalifa, artillery, Taiasha.
omdurman-rules/src/board.rs:311 GH:omdurman-rules/src/board.rs:311	**is_walled_city**	/// are its seeds). The set is derived once from the board data, replacing /// the older "at least two of six hexsides are Wall/Gate/Breach" heuristic. pub fn is_walled_city(&self, hex: HexCoord) -> bool { // Membership in the precomputed enclosed area (see `walled_city`). // Palace/Tomb hexes are always part of it (they are the seeds).
omdurman-rules/src/effects/error.rs:90 GH:omdurman-rules/src/effects/error.rs:90	**WalledCityEntry**	 #[error("unit {0:?} is not eligible to enter the walled city of Omdurman at {1:?} (§5.23)")] WalledCityEntry(UnitId, HexCoord), #[error("movement cost {cost:?} exceeds allowance {allowance:?}")]

Proven by: **omdurman-types::src::lib::hexside_ref_is_order_independent**

Covered by tests: **omdurman-rules::src::effects::tests::can_move_rejects_wall_hexside** **omdurman-rules::src::effects::tests::can_move_allows_gate_hexside** **omdurman-rules::src::effects::tests::walled_city_entry_allows_khalifa** **omdurman-rules::src::effects::tests::walled_city_entry_rejects_unauthorized_dervish** **omdurman-rules::src::effects::tests::walled_city_entry_rejects_ae_gunboat** **omdurman-rules::src::effects::tests::walled_city_entry_not_enforced_for_fok** **omdurman-rules::src::board_data::campaign_walled_city_is_enclosed_by_walls**

§5.24 – Gunboat upstream/downstream movement

implemented

manual page unknown

> Note that gunboats have two movement allowances separated by a slash, e.g. 10/16. The smaller number is the movement allowance when moving upstream, i.e. against the current (the direction of the current is indicated by arrows in the Nile). The larger number is the movement allowance when moving downstream, i.e. with the current. Gunboats may combine movement in both directions, but if they move even one hex upstream, their upstream movement allowance is their maximum movement allowance for that turn, and may not be exceeded.

(see manual for full text)

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:552 GH:omdurman-rules/src/lib.rs:552	**GunboatMovement**	/// the turn. #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] pub struct GunboatMovement {

		`553 │     pub upstream: MovementAllowance,` `554 │     pub downstream: MovementAllowance,`
omdurman-types/src/lib.rs:956 GH:omdurman-types/src/lib.rs:956	**is_boat**	`954 │` `955 │     /// Gunboats use the split upstream/downstream movement allowance (§5.24).` `956 │     pub fn is_boat(self) -> bool {` `957 │         matches!(self, UnitKind::Gunboat { .. })` `958 │     }`

Covered by tests: **omdurman-rules::src::unit_profiles::boat_annotation_yields_split_gunboat_movement** **omdurman-rules::src::effects::tests::gunboat_upstream_cap_is_sticky_across_moves**

§5.25 – Dervish forts may not move

implemented

manual page unknown

> Dervish forts may not move in any way once placed.

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:813 GH:omdurman-rules/src/lib.rs:813	**Immobile**	`811 │     Gunboat(GunboatMovement),` `812 │     /// Forts may not move once placed (§5.25).` `813 │     Immobile,` `814 │ }` `815 │`
omdurman-types/src/lib.rs:913 GH:omdurman-types/src/lib.rs:913	**UnitKind::Fort**	`911 │     /// Permanent emplacement (§6.54): fire (artillery) / melee (defensive).` `912 │     /// May not move once placed (§5.25).` `913 │     Fort { fire: i32, melee: i32 },` `914 │     /// Dervish leader (§6.51): fire / melee / movement. May melee attack (§7.4).` `915 │     DervishLeader {`
omdurman-rules/src/lib.rs:813 GH:omdurman-rules/src/lib.rs:813	**UnitMovement::Immobile**	`811 │     Gunboat(GunboatMovement),` `812 │     /// Forts may not move once placed (§5.25).` `813 │     Immobile,` `814 │ }` `815 │`

Covered by tests: **omdurman-rules::src::effects::tests::forts_are_never_advance_eligible**

§5.26 – Units stop on entering enemy ZOC

implemented

manual page unknown

> Units must stop their movement immediately upon entering an enemy zone of control (see 5.4).

See also: §5.4

File	Symbol	Code Snippet
omdurman-rules/src/effects/state.rs:682 GH:omdurman-rules/src/effects/state.rs:682	**can_move_unit_to**	`680 │     ///` ``681 │     /// [`hex_in_enemy_zoc`]: Self::hex_in_enemy_zoc`` `682 │     pub fn can_move_unit_to(` `683 │         &self,` `684 │         unit_id: UnitId,`
omdurman-rules/src/effects/	**hex_in_enemy_zoc**	`1480 │     /// does not extend into or out of a Nile hex. With no board loaded these`

state.rs:1482 GH:omdurman-rules/src/effects/state.rs:1482		`1481 |     /// reduce to the plain adjacency rule.` `1482 |     pub fn hex_in_enemy_zoc(` `1483 |         &self,` `1484 |         hex: HexCoord,`

Covered by tests: **omdurman-rules::src::effects::tests::unit_entering_enemy_zoc_may_move_no_further_that_turn** **omdurman-rules::src::effects::tests::zoc_transit_check_uses_the_actual_path**

§5.41 – All units except AE leaders exert ZOC

implemented

manual page unknown

> All units except Anglo-Egyptian leaders exert a zone of control (hereafter called a ZOC) into their six adjacent hexes (exception: Gunboats exert a ZOC only against enemy gunboats). Disrupted units have no ZOC.

File	Symbol	Code Snippet
omdurman-rules/src/effects/state.rs:1729 GH:omdurman-rules/src/effects/state.rs:1729	**unit_projects_zoc_rule**	`1727 | /// * A fort projects ZOC out of its hex even when unoccupied (§5.44),` `1728 | ///   modelled by the fort unit projecting normally.` `1729 | pub fn unit_projects_zoc_rule(` `1730 |     unit: &UnitPlacement,` `1731 |     mover_player: Player,`
omdurman-rules/src/lib.rs:868 GH:omdurman-rules/src/lib.rs:868	**ZocReason**	`866 | /// Used by the engine when answering "is this hex in an enemy ZOC?".` `867 | #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)]` `868 | pub enum ZocReason {` `869 |     /// Normal ZOC: any non-disrupted unit other than an Anglo-Egyptian` `870 |     /// leader (§5.41) projects ZOC into each of its six adjacent hexes.`
omdurman-rules/src/effects/state.rs:1464 GH:omdurman-rules/src/effects/state.rs:1464	**unit_projects_zoc**	`1462 |     /// §5.44) need the game map, which the engine does not hold; the app layers` `1463 |     /// those on top. This is the position/kind/disruption core of the rule.` `1464 |     pub fn unit_projects_zoc(` `1465 |         &self,` `1466 |         unit: &UnitPlacement,`
omdurman-rules/src/effects/state.rs:1523 GH:omdurman-rules/src/effects/state.rs:1523	**zoc_hexes**	`1521 |     /// ZOC covers a given hex; this function returns *which* hexes a` `1522 |     /// specific unit covers.` `1523 |     pub fn zoc_hexes(` `1524 |         &self,` `1525 |         unit: &UnitPlacement,`

Proven by: **omdurman-rules::src::effects::unit_projects_zoc_matches_manual_clauses**

Covered by tests: **omdurman-rules::src::effects::tests::zoc_hexes_empty_for_anglo_egyptian_leader** **omdurman-rules::src::effects::tests::zoc_hexes_normal_unit_projects_six_adjacent_minus_exclusions** **omdurman-rules::src::effects::tests::zoc_hexes_empty_for_disrupted_unit** **omdurman-rules::src::effects::tests::zoc_hexes_matches_hex_in_enemy_zoc** **omdurman-rules::src::effects::tests::zoc_hexes_excludes_nile** **omdurman-rules::src::effects::tests::zoc_hexes_excludes_khor**

§5.42 – No MP cost to enter/leave enemy ZOC

implemented

manual page unknown

> There is no movement point cost to enter or leave an enemy ZOC.

File	Symbol	Code Snippet
omdurman-rules/src/effects/state.rs:848 GH:omdurman-rules/src/effects/state.rs:848	**movement_cost_for**	`    ///` `    /// §5.42: entering or leaving an enemy ZOC adds no MP cost.` `    pub fn movement_cost_for(` `        &self,` `        unit: &UnitPlacement,`

Covered by tests: **omdurman-rules::src::effects::tests::entering_enemy_zoc_costs_no_extra_mp**

§5.43 – Units stop when entering enemy ZOC

implemented

manual page unknown

> All units must stop when they enter an enemy ZOC and may move no further that turn. In their next movement phase they may withdraw or, if desired, move directly into another enemy ZOC.

File	Symbol	Code Snippet
omdurman-rules/src/effects/state.rs:682 GH:omdurman-rules/src/effects/state.rs:682	**can_move_unit_to**	`    ///` `    /// [`hex_in_enemy_zoc`]: Self::hex_in_enemy_zoc` `    pub fn can_move_unit_to(` `        &self,` `        unit_id: UnitId,`
omdurman-rules/src/effects/state.rs:1482 GH:omdurman-rules/src/effects/state.rs:1482	**hex_in_enemy_zoc**	`    /// does not extend into or out of a Nile hex. With no board loaded these` `    /// reduce to the plain adjacency rule.` `    pub fn hex_in_enemy_zoc(` `        &self,` `        hex: HexCoord,`

Covered by tests: **omdurman-rules::src::effects::tests::unit_entering_enemy_zoc_may_move_no_further_that_turn**

§5.44 – ZOC limitations (walls, khor, fort, Nile, Zariba)

implemented

manual page unknown

> ZOCs do not extend into or out of a Nile River hex (exception: Gunboats, see 5.41). ZOCs do not extend across a khor, into a fort, or into a hex inside the walled city across a wall hexside. ZOCs do extend out of a fort (even if unoccupied), and from a walled city hex into an adjacent non-walled-city hex across a wall hexside. ZOCs also extend out of (but not into) a walled city hex across a gate hexside. ZOCs extend both ways across a breach hexside. ZOCs also extend out of, but not into, a hut or building hex. In the historical scenario ZOCs extend out of, but not into, the Zariba across a Zariba hexside (also in the campaign game if the Zariba is constructed).

(see manual for full text)

See also: §5.41

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:868	**ZocReason**	`/// Used by the engine when answering "is this hex in an enemy ZOC?".` `#[derive(Serialize, Deserialize, Clone, Copy,`

GH:omdurman-rules/src/lib.rs:868		`PartialEq, Eq, Debug)]` `868  pub enum ZocReason {` `869      /// Normal ZOC: any non-disrupted unit other than an Anglo-Egyptian` `870      /// leader (§5.41) projects ZOC into each of its six adjacent hexes.`
omdurman-types/src/lib.rs:192 GH:omdurman-types/src/lib.rs:192	**Wall**	`190      /// (§5.44), blocks melee (§7.2), blocks advance-after-combat (§6.82).` `191      #[default]` `192      Wall,` `193      /// Gate hexside in a wall. ZOC extends *out of* the walled city through` `194      /// gates but not into it (§5.44). Melee may be made through a gate (§7.2).`
omdurman-types/src/lib.rs:201 GH:omdurman-types/src/lib.rs:201	**Khor**	`199      /// Khor -- gully/wadi. ZOCs do not extend across (§5.44); advance after` `200      /// combat may not cross (§6.82).` `201      Khor,` `202      /// Crest line. Blocks LOS unless the firer is on the higher side` `203      /// (§6.3 note 7).`
omdurman-rules/src/lib.rs:881 GH:omdurman-rules/src/lib.rs:881	**ZocReason::Zariba**	`879      /// Zariba hexside ZOC behaviour in the historical scenario / when the` `880      /// Zariba is constructed (§5.44).` `881      Zariba,` `882  }` `883`
omdurman-rules/src/effects/state.rs:1464 GH:omdurman-rules/src/effects/state.rs:1464	**unit_projects_zoc**	`1462      /// §5.44) need the game map, which the engine does not hold; the app layers` `1463      /// those on top. This is the position/kind/disruption core of the rule.` `1464      pub fn unit_projects_zoc(` `1465          &self,` `1466          unit: &UnitPlacement,`
omdurman-types/src/lib.rs:275 GH:omdurman-types/src/lib.rs:275	**HexsideKind::blocks_zoc**	`273      /// cannot express; those are left to the caller. This predicate captures the` `274      /// symmetric "does not extend across" cases.` `275      pub fn blocks_zoc(self) -> bool {` `276          matches!(` `277              self,`
omdurman-rules/src/effects/state.rs:1482 GH:omdurman-rules/src/effects/state.rs:1482	**hex_in_enemy_zoc**	`1480      /// does not extend into or out of a Nile hex. With no board loaded these` `1481      /// reduce to the plain adjacency rule.` `1482      pub fn hex_in_enemy_zoc(` `1483          &self,` `1484          hex: HexCoord,`

Proven by: **omdurman-rules::src::effects::unit_projects_zoc_matches_manual_clauses**

Covered by tests: **omdurman-rules::src::effects::tests::zoc_hexes_excludes_nile** **omdurman-rules::src::effects::tests::zoc_hexes_excludes_khor** **omdurman-rules::src::effects::tests::zoc_hexes_normal_unit_projects_six_adjacent_minus_exclusions**

§5.51 – Stacking limit (4 units + leaders, gunboats isolated)

implemented

manual page unknown

> No more than four units may occupy a hex, with the exception of leaders and gunboats. All leader units are free stacking, i.e. they may stack in addition to the four-unit-per-hex stacking limitation. Gunboats may not stack with any other unit (Exception: 5.21). Players may move through friendly units at no additional cost in movement points. The stacking limitation applies only at the end of the movement phase and during combat.

See also: §5.21

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:893 GH:omdurman-rules/src/lib.rs:893	OverLimit	`891 │     /// (§6.51: a Dervish unit occupying his hex eliminates him).` `892 │     #[error("hex stack exceeds the four-unit limit [§5.51]")]` `893 │     OverLimit,` `894 │     /// "Gunboats may not stack with any other unit" (§5.51, exception §5.21).` `895 │     #[error("gunboats may not stack with non-gunboat units [§5.51]")]`
omdurman-rules/src/lib.rs:896 GH:omdurman-rules/src/lib.rs:896	GunboatStack	`894 │     /// "Gunboats may not stack with any other unit" (§5.51, exception §5.21).` `895 │     #[error("gunboats may not stack with non-gunboat units [§5.51]")]` `896 │     GunboatStack,` `897 │     /// "Units of different Dervish tribes may not stack together" (§5.52).` `898 │     /// The Dervish artillery (§9.322's three guns) is not a tribe but is not`
omdurman-rules/src/lib.rs:906 GH:omdurman-rules/src/lib.rs:906	EnemyCohabitation	`904 │     /// lone Anglo-Egyptian leader is exempt (§6.51).` `905 │     #[error("enemy units may not share a hex; melee, not movement, engages them [§5.51, §7.1]")]` `906 │     EnemyCohabitation,` `907 │     /// "If Dervish leaders elect to stack, they may only stack with units of` `908 │     /// their command (i.e. colour)" (§5.53).`
omdurman-rules/src/effects/state.rs:1415 GH:omdurman-rules/src/effects/state.rs:1415	check_stacking	`1413 │     ///   Dervish artillery is its own group) may not stack together.` `1414 │     /// * §5.53 -- a Dervish leader may stack only with units of its command.` `1415 │     pub fn check_stacking(` `1416 │         &self,` `1417 │         mover: &UnitPlacement,`
omdurman-rules/src/effects/state.rs:1640 GH:omdurman-rules/src/effects/state.rs:1640	stacking_rule	`1638 │ ///   artillery as its own group) may not share a hex.` `1639 │ /// * §5.53: a Dervish leader stacks only with units of its command.` `1640 │ pub fn stacking_rule(occupants: &[&UnitPlacement]) -> Result<(), crate::StackingError> {` `1641 │     use crate::StackingError;` `1642 │`

Proven by: omdurman-rules::src::effects::stacking_rule_cohabitation_is_exact omdurman-rules::src::effects::stacking_rule_is_symmetric omdurman-rules::src::effects::stacking_rule_leaders_are_free_stacking omdurman-rules::src::effects::stacking_rule_limit_is_four_counted_units omdurman-rules::src::effects::stacking_rule_gunboat_never_shares

Covered by tests: omdurman-rules::src::effects::tests::stacking_over_limit_rejected omdurman-rules::src::effects::tests::mid_move_stacking_allows_pass_through omdurman-rules::src::effects::tests::mid_move_stacking_rejects_over_limit_destination omdurman-rules::src::effects::tests::validate_stacking_invariants_clean_state omdurman-rules::src::effects::tests::validate_stacking_invariants_catches_stacking_violation omdurman-rules::src::effects::tests::validate_stacking_invariants_allows_leaders_stacking omdurman-rules::src::effects::tests::deploy_rejects_enemy_cohabitation_during_setup omdurman-rules::src::effects::tests::validate_stacking_invariants_catches_enemy_cohabitation omdurman-rules::src::effects::tests::ae_garrison_stacks_freely_under_gordon omdurman-rules::src::effects::tests::melee_mandatory_advance_leaders_are_free_stacking

§5.52 – Different Dervish tribes may not stack together

implemented

manual page unknown

> The units of different Dervish tribes may not stack together, even if they are the same color (e.g. although both are green, Mulazmin and Jehadia units may not stack with each other).

<table>
<tr><th>File</th><th>Symbol</th><th>Code Snippet</th></tr>
<tr><td>omdurman-rules/src/lib.rs:902
GH:omdurman-rules/src/lib.rs:902</td><td>DervishTribeMix</td><td><pre>900 | /// [`UnitIdentity::dervish_stacking_group`]).
901 | #[error("Dervish units of different tribes may not stack [§5.52]")]
902 | DervishTribeMix,
903 | /// A unit may never share a hex with enemy units (§5.51, §7.1); only the
904 | /// lone Anglo-Egyptian leader is exempt (§6.51).</pre></td></tr>
<tr><td>omdurman-rules/src/lib.rs:707
GH:omdurman-rules/src/lib.rs:707</td><td>dervish_stacking_group</td><td><pre>705 | /// `None` for every unit the §5.52 law does not constrain: Dervish
706 | /// leaders, forts and gunboats, and all Anglo-Egyptian units.
707 | pub fn dervish_stacking_group(&self) -> Option<DervishStackingGroup> {
708 | match self {
709 | UnitIdentity::DervishTribal { tribe } => Some(DervishStackingGroup::Tribe(*tribe)),</pre></td></tr>
</table>

Proven by: **omdurman-rules::src::effects::stacking_rule_group_purity_is_exact**

Covered by tests: **omdurman-rules::src::unit_profiles::green_sections_are_mulazmin_tribal_units** **omdurman-rules::src::effects::tests::deploy_rejects_dervish_tribe_mix** **omdurman-rules::src::effects::tests::deploy_rejects_hadendowa_on_dervish_gun** **omdurman-rules::src::effects::tests::dervish_guns_stack_with_guns_and_their_leader** **omdurman-rules::src::effects::tests::validate_stacking_invariants_clean_state**

§5.53 – Leader stacking with command colour only

implemented

manual page unknown

> Leader units are not required to stack. If Dervish leaders elect to stack, however, they may only stack with units of their command (i.e. color). For example, Sheik El Din may only stack with Mulazmins or Jehadias.
>
> **5.54) Anglo-Egyptian Brigade Integrity:** All British, Sudanese, and Egyptian infantry units have their brigade designation printed in the upper right corner (e.g. "2B" = 2nd British Brigade; "3E" = 3rd Egyptian Brigade, etc.). In any combat phase in which all four infantry battalions belonging to any Anglo-Egyptian infantry brigade are stacked in the same hex they are said to have brigade integrity. Stacks having brigade integrity receive a +1 modifier to their fire combat die roll provided they all fire at the same enemy occupied hex. This modifier is in addition to the normal +1 bonus given to all Anglo-Egyptian direct fire attacks (see 6.24).

(see manual for full text)

See also: §5.54, §6.24

<table>
<tr><th>File</th><th>Symbol</th><th>Code Snippet</th></tr>
<tr><td>omdurman-rules/src/lib.rs:910
GH:omdurman-rules/src/lib.rs:910</td><td>DervishLeaderCommandMismatch</td><td><pre>908 | /// their command (i.e. colour)" (§5.53).
909 | #[error("Dervish leader may only stack with units of their own command [§5.53]")]
910 | DervishLeaderCommandMismatch,
911 | }
912 |</pre></td></tr>
</table>

Proven by: **omdurman-rules::src::effects::stacking_rule_leaders_are_free_stacking**

Covered by tests: **omdurman-rules::src::effects::tests::dervish_leader_stacks_only_with_command_colour**

§5.54 – Anglo-Egyptian Brigade Integrity

implemented

manual page unknown

<table>
<tr><th>File</th><th>Symbol</th><th>Code Snippet</th></tr>
<tr><td>omdurman-rules/src/lib.rs:916
GH:omdurman-rules/src/lib.rs:916</td><td>BrigadeIntegrity</td><td><pre>914 | /// stack contains all four battalions of a single Anglo-Egyptian brigade.
915 | #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)]
916 | pub enum BrigadeIntegrity {
917 | None,
918 | Integrated(BrigadeId),</pre></td></tr>
<tr><td>omdurman-rules/src/lib.rs:771
GH:omdurman-rules/src/lib.rs:771</td><td>brigade_integrity</td><td><pre>769 | /// Only a full stack of four battalions qualifies. Three or fewer may still
770 | /// stack and fire, but they receive no brigade-integrity bonus.
771 | pub fn brigade_integrity(identities: &[UnitIdentity]) -> BrigadeIntegrity {
772 | let Some(brigade) = identities.first().and_then(|i| i.brigade()) else {
773 | return BrigadeIntegrity::None;</pre></td></tr>
<tr><td>omdurman-rules/src/lib.rs:916
GH:omdurman-rules/src/lib.rs:916</td><td>FireModifier::BrigadeIntegrity</td><td><pre>914 | /// stack contains all four battalions of a single Anglo-Egyptian brigade.
915 | #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)]
916 | pub enum BrigadeIntegrity {
917 | None,
918 | Integrated(BrigadeId),</pre></td></tr>
<tr><td>omdurman-rules/src/lib.rs:329
GH:omdurman-rules/src/lib.rs:329</td><td>BattalionOrdinal</td><td><pre>327 | /// brigade integrity requires all four stacked in one hex (§5.54).
328 | #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)]
329 | pub enum BattalionOrdinal {
330 | First = 1,
331 | Second = 2,</pre></td></tr>
<tr><td>omdurman-types/src/lib.rs:1021
GH:omdurman-types/src/lib.rs:1021</td><td>BrigadeId</td><td><pre>1019 | /// same field for uniform handling.
1020 | #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug)]
1021 | pub struct BrigadeId {
1022 | pub number: u8,
1023 | pub nationality: BrigadeNationality,</pre></td></tr>
</table>

Proven by: **omdurman-rules::src::lib::brigade_integrity_requires_all_four_distinct_battalions_of_one_brigade**

Covered by tests: **omdurman-rules::src::unit_profiles::brigade_designation_ignored_for_non_infantry** **omdurman-rules::src::unit_profiles::printed_brigade_designation_overrides_column** **omdurman-rules::src::unit_profiles::section_owner_anglo_egyptian_sections** **omdurman-rules::src::lib::brigade_integrity_four_battalions_returns_integrated** **omdurman-rules::src::unit_profiles::ae_infantry_fourth_battalion_from_col_3** **omdurman-rules::src::lib::brigade_integrity_empty_slice** **omdurman-rules::src::lib::brigade_integrity_friendlies_returns_none** **omdurman-rules::src::lib::brigade_integrity_three_battalions_returns_none** **omdurman-rules::src::lib::unit_identity_brigade_and_battalion_accessors** **omdurman-rules::src::unit_profiles::ae_infantry_brigade_number_three_from_col_7** **omdurman-rules::src::lib::brigade_integrity_non_infantry_returns_none** **omdurman-rules::src::lib::brigade_integrity_mixed_brigades_returns_none** **omdurman-rules::src::unit_profiles::ae_infantry_third_battalion_from_col_2** **omdurman-rules::src::effects::tests::brigade_integrity_modifier_is_engine_derived**

21/31 implemented

§6 – Fire Combat Phase

§6 – Fire Combat Phase

descriptive

manual page unknown

> Fire Combat Phase

§6.1 – General Rules

descriptive

manual page unknown

> General Rules

§6.2 – How To Have Fire Combat

descriptive

manual page unknown

> How To Have Fire Combat

§6.3 – Line of Sight Table

implemented

manual page unknown

> Line of Sight Table
>
> This table is located on the back of this rulebook and should be self-explanatory. Locate the terrain type the firing unit is in and cross-index it with the terrain type the target unit is in. Terrain types in the intersecting box block line of sight, with exceptions as footnoted. Also study the “Special LOS Notes” given and remember that (with the exception of howitzer fire — see 6.64) you can’t fire at anything you can’t see!

See also: §6.64

File	Symbol	Code Snippet
omdurman-rules/src/los_table.rs:57 GH:omdurman-rules/src/los_table.rs:57	**LosLevel**	`    serde::Serialize, serde::Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Hash, Debug,` `)]` `pub enum LosLevel {` `    Ground,` `    Rough,`
omdurman-rules/src/los_table.rs:81 GH:omdurman-rules/src/los_table.rs:81	**LosFeature**	`/// authored RON spellings.` `#[derive(serde::Serialize, serde::Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug)]` `pub enum LosFeature {` `    /// A hex containing units (gunboats/forts excluded per note a).` `    Units,`
omdurman-rules/src/los_table.rs:103 GH:omdurman-rules/src/los_table.rs:103	**LosCondition**	`/// A positional condition from the LOS table Detail footnotes.` `#[derive(serde::Serialize, serde::Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug)]` `pub enum LosCondition {` `    /// (1) Blocks only if the ray passes through more`

		`than two such features.` `105 │     MoreThanTwo,`
omdurman-rules/src/los_table.rs:183 GH:omdurman-rules/src/los_table.rs:183	**los_level**	`181 │ ///` `182 │ /// For all other units, the level is derived from the terrain at `hex`.` `183 │ pub fn los_level_for_unit(` `184 │     kind: UnitKind,` `185 │     hex: HexCoord,`
omdurman-rules/src/los_table.rs:183 GH:omdurman-rules/src/los_table.rs:183	**los_level_for_unit**	`181 │ ///` `182 │ /// For all other units, the level is derived from the terrain at `hex`.` `183 │ pub fn los_level_for_unit(` `184 │     kind: UnitKind,` `185 │     hex: HexCoord,`
omdurman-rules/src/los_table.rs:223 GH:omdurman-rules/src/los_table.rs:223	**blocking_rules**	`221 │ /// always blocks. Indexing is in-bounds by construction (both enums have` `222 │ /// exactly three variants).` `223 │ pub fn blocking_rules(firer: LosLevel, target: LosLevel) -> &'static [BlockingRule] {` `224 │     crate::tables_data::LOS_CELLS[firer.index()][target.index()]` `225 │ }`
omdurman-rules/src/los_table.rs:303 GH:omdurman-rules/src/los_table.rs:303	**has_los**	`301 │ /// `unit_level_at` closure returns the LOS level of blocking units` `302 │ /// (non-gunboat, non-fort per note a) in an intervening hex, or `None`.` `303 │ pub fn has_los(` `304 │     board: &crate::board::BoardInfo,` `305 │     from: HexCoord,`
omdurman-types/src/lib.rs:230 GH:omdurman-types/src/lib.rs:230	**HexsideKind::blocks_los**	`228 │     /// (LOS table conditions 2–4, 7) and note (e) are handled by the engine` `229 │     /// in `omdurman_rules::los_table`, not by this predicate.` `230 │     pub fn blocks_los(self) -> bool {` `231 │         matches!(self, HexsideKind::Wall | HexsideKind::Crest)` `232 │     }`
omdurman-types/src/lib.rs:230 GH:omdurman-types/src/lib.rs:230	**Terrain::blocks_los**	`228 │     /// (LOS table conditions 2–4, 7) and note (e) are handled by the engine` `229 │     /// in `omdurman_rules::los_table`, not by this predicate.` `230 │     pub fn blocks_los(self) -> bool {` `231 │         matches!(self, HexsideKind::Wall | HexsideKind::Crest)` `232 │     }`
omdurman-types/src/lib.rs:452 GH:omdurman-types/src/lib.rs:452	**Terrain::is_los_trees**	`450 │     /// (§6.3 note 1). Retained for compatibility; the full LOS engine` `451 │     /// checks `Terrain::Trees` directly.` `452 │     pub fn is_los_trees(self) -> bool {` `453 │         matches!(self, Terrain::Trees { .. })` `454 │     }`

Proven by: **omdurman-types::src::lib::line_between_forms_a_connected_ray**

Covered by tests: **omdurman-rules::src::los_table::los_level_mapping** **omdurman-rules::src::los_table::blocking_rules_all_cells_covered** **omdurman-rules::src::los_table::has_los_empty_board_is_clear** **omdurman-rules::src::los_table::has_los_adjacent_clear** **omdurman-rules::src::los_table::has_los_howitzer_bypasses** **omdurman-rules::src::los_table::has_los_wall_hexside_blocks_ground_to_ground** **omdurman-rules::src::los_table::has_los_gate_hexside_passes** **omdurman-rules::src::los_table::has_los_breach_hexside_passes** **omdurman-rules::src::los_table::has_los_rough_intervening_blocks_ground_to_ground** **omdurman-rules::src::los_table::has_los_two_tree_hexes_pass_ground_to_ground** **omdurman-rules::src::los_table::has_los_three_tree_hexes_block_ground_to_ground** **omdurman-rules::src::los_table::has_los_two_hut_hexes_pass_ground_to_ground** **omdurman-rules::src::los_table::has_los_three_hut_hexes_block_ground_to_ground**

omdurman-rules::src::los_table::has_los_hilltop_to_hilltop_clear_no_units
omdurman-rules::src::los_table::has_los_hilltop_to_hilltop_blocked_by_hilltop_unit
omdurman-rules::src::los_table::has_los_hilltop_to_hilltop_not_blocked_by_ground_unit
omdurman-rules::src::los_table::has_los_rough_to_rough_unit_at_lower_level_passes
omdurman-rules::src::los_table::has_los_rough_to_rough_unit_at_same_level_blocks
omdurman-rules::src::los_table::has_los_rough_to_rough_hilltop_blocks
omdurman-rules::src::los_table::has_los_ground_to_hilltop_intervening_hilltop_blocks
omdurman-rules::src::los_table::has_los_building_blocks_like_huts_ground_to_ground
omdurman-rules::src::los_table::has_los_two_building_hexes_pass_ground_to_ground
omdurman-rules::src::los_table::los_level_for_unit_gunboat_is_rough
omdurman-rules::src::los_table::los_level_for_unit_fort_is_ground
omdurman-rules::src::los_table::los_level_for_unit_walled_city_adj_wall_is_rough
omdurman-rules::src::los_table::gunboat_firer_uses_rough_row_not_ground
omdurman-rules::src::los_table::los_reflexive_all_terrains
omdurman-rules::src::los_table::los_reflexive_hilltop omdurman-rules::src::los_table::los_symmetric_ground_to_ground_no_units
omdurman-rules::src::los_table::los_howitzer_always_has_los
omdurman-rules::src::los_table::los_howitzer_same_hex omdurman-rules::src::los_table::los_blocking_rules_match_reference_table
omdurman-rules::src::los_table::los_ground_to_ground_features_block_as_expected

§6.4 – Fire Combat Sequence

implemented

manual page unknown

> Fire Combat Sequence
>
> The sequence of fire combat resolution is the same for both defensive and offensive fire combat. During the Dervish player turn the Anglo-Egyptian player executes 6.41 AND 6.42 as defensive fire, after which the Dervish player executes 6.41 as offensive fire. During the Anglo-Egyptian player turn the Dervish player executes 6.41 as defensive fire, after which the Anglo-Egyptian player executes 6.41 AND 6.42 as offensive fire.
>
> **6.41) Direct Fire Subphase (Dervish and Anglo-Egyptian players):** The firing player must first allocate all of his fire attacks, combining his units' direct fire combat factors in any manner he wishes. After all fire has been allocated, the firing player then resolves his attacks in any order he wishes.
>
> **6.42) Maxim Second Fire and Howitzer Fire Subphase (Anglo-Egyptian player only):** Anglo-Egyptian named gunboats may now fire their artillery factor as howitzer fire (see 6.64) and all Maxim guns may fire a second time. Once

(see manual for full text)

See also: §6.41, §6.42, §6.64

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:302 GH:omdurman-rules/src/lib.rs:302	FireSubPhase	`300 │` `301 │ #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)]` `302 │ pub enum FireSubPhase {` `303 │     /// Direct fire (§6.41). Both sides participate in this sub-phase.` `304 │     DirectFire,`
omdurman-rules/src/effects/effect.rs:72 GH:omdurman-rules/src/effects/effect.rs:72	FireCombat	` 70 │     /// - Firers marked as fired; target hex marked as fired-at.` ` 71 │     /// - Victory points awarded for eliminations.` ` 72 │     FireCombat { attack: FireAttack, roll: DieRoll },` ` 73 │` ` 74 │     /// Resolve a howitzer bombardment (two rolls: CRT + impact scatter)`

Covered by tests: omdurman-rules::src::effects::tests::turn_advances_through_phases

§6.5 – Special Unit Capabilities

descriptive

manual page unknown

> Special Unit Capabilities
>
> **6.51) Leader Units:** Dervish leader units have a fire factor, a melee factor, and a movement factor. They may thus attack, melee, and be eliminated like any other combat unit. Their special benefit is that they stack free.
>
> Anglo-Egyptian leaders have a movement factor only. They are eliminated if a) they are alone in a hex when a Dervish unit occupies or passes through that hex, or b) if all of the combat units a leader is stacked with are eliminated in fire combat or melee. The special function of Anglo-Egyptian leaders is that at least one must survive to occupy the Mahdi's tomb hex if it is to be taken from the Dervish player (see 9.14).
>
> **6.52) Anglo-Egyptian "Friendlies" Brigade:** These units represent native volunteers in the Anglo-Egyptian army. They fire rifles on the Dervish Range Effects Table and melee with the Dervish melee modifier. They may not enter the walled city of Omdurman (see 5.23). They may be transferred to the west bank (see 5.21).

(see manual for full text)

See also: §5.21, §5.23, §5.44, §6.51, §6.52, §6.53, §6.54, §6.62, §6.63, §7.6, §9.14

§6.6 – Special Artillery Capabilities

descriptive

manual page unknown

> Special Artillery Capabilities

§6.7 – Defensive Fire

implemented

manual page unknown

> Defensive Fire
>
> In Defensive Fire phase, all of the non-moving player's units may fire at any of the moving player's units in range, within the limitations imposed by the rules of combat (see 6.1 to 6.6). There is no advance after combat as a result of defensive fires.

See also: §6.1, §6.6

File	Symbol	Code Snippet
omdurman-rules/src/effects/state.rs:1919 GH:omdurman-rules/src/effects/state.rs:1919	can_advance_after_comb	`1917 │     /// player's unit, not artillery, adjacent to `to`, and `to` now empty.` `1918 │     /// Wall/khor hexside restrictions are not enforced (no hexside map data).` `1919 │     pub fn can_advance_after_combat(&self, unit_id: UnitId, to: HexCoord) -> Result<(), RuleError> {` `1920 │         let unit = self.unit_or_err(unit_id)?;` `1921 │         // §6.7: there is no advance after combat as a result of defensive fire.`

Covered by tests: omdurman-rules::src::effects::tests::no_advance_after_defensive_fire omdurman-rules::src::effects::tests::defensive_fire_opens_no_advance_window

§6.8 – Offensive Fire

descriptive

manual page unknown

> Offensive Fire

§6.11 – Fire combat factor printed on units

implemented

manual page unknown

> The fire combat factor of the various unit types is printed directly on the units and is a numerical expression of the unit's fire strength.

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:86 GH:omdurman-rules/src/lib.rs:86	**FireFactor**	`    /// Every possible value from the annotated counter set is a named variant.` `    #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)]` `    pub enum FireFactor {` `        One = 1,` `        Three = 3,`

Covered by tests: **omdurman-rules::src::lib::fire_factor_sum_to_row**

§6.12 – Fire combat is always voluntary

implicit

manual page unknown

> Fire combat is always voluntary. A unit is never required to fire at enemy units merely because they are in range or adjacent.

§6.13 – Fire factor is unitary (may not be divided)

implicit

manual page unknown

> If a unit elects to fire, its fire combat factor at an enemy unit, that fire combat factor is unitary. A unit's fire combat factor may not be divided up to fire at enemy units on different hexes.

§6.14 – Players may combine fire factors into one attack

implemented

manual page unknown

> Players may combine fire during fire combat phase, i.e. they may fire at an enemy-occupied hex with as many friendly units as may legally do so, combining all of their fire combat factors into one attack. Note that in any given fire combat phase, however, a combat unit may only fire once and may only be fired at once (exceptions: Maxim guns and gunboats — see 6.4).

See also: §6.4

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:100 GH:omdurman-rules/src/lib.rs:100	**sum_to_row**	`impl FireFactor {` `    /// Sum multiple fire factors and return the corresponding Combat Results Table row (rulebook §6.11).` `    pub fn sum_to_row<'a>(factors: impl IntoIterator<Item = &'a FireFactor>) -> FireFactorRow {` `        let total: u16 = factors.into_iter().map(|f| f.value()).sum();` `crate::combat_results_table::FireFactorRow::from_total(total)`

Covered by tests: **omdurman-rules::src::effects::tests::unit_may_only_be_fired_at_once_per_phase** **omdurman-rules::src::effects::tests::gunboat_and_maxim_may_be_fired_at_repeatedly** **omdurman-rules::src::effects::tests::rejected_fire_attack_does_not_mark_firers_as_fired**

§6.15 – May divide a stack to fire at different hexes

implicit

manual page unknown

> Players may also divide a stack of units in order to fire at different enemy-occupied hexes. Anglo-Egyptian infantry units having brigade integrity, however, do not receive their +1 direct fire modifier unless they all fire at the same enemy-occupied hex (see 5.54).

See also: §5.54

§6.16 – Halving fire strength rounds down, minimum 1

implicit

implemented

manual page unknown

> When halving fire combat strength, always round down each individual unit. For example, an Egyptian brigade of four battalions, each having a printed strength of 9 fire factors, will fire a total of 16 factors when halved. However, a unit's firing strength is never reduced below one by halving.

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:519 GH:omdurman-rules/src/lib.rs:519	**RangeBand**	`517 │ /// multiplied at a given distance (§6.22).` `518 │ #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)]` `519 │ pub enum RangeBand {` `520 │     Tripled,` `521 │     Doubled,`

Proven by: **omdurman-rules::src::lib::range_band_halved_is_max_of_one_and_floor_half** **omdurman-rules::src::lib::range_band_apply_is_monotone_in_raw** **omdurman-rules::src::lib::disrupt_half_is_rounded_up**

Covered by tests: **omdurman-rules::src::lib::halving_rounds_down_and_never_below_one**

§6.21 – First check LOS before firing

implemented

manual page unknown

> When combat units wish to fire at enemy units, first check the Line of Sight Table to be sure the firing unit can see the target hex (exception: howitzer fire, see 6.64).

See also: §6.64

File	Symbol	Code Snippet
omdurman-rules/src/los_table.rs:223 GH:omdurman-rules/src/los_table.rs:223	**blocking_rules**	`221 │ /// always blocks. Indexing is in-bounds by construction (both enums have` `222 │ /// exactly three variants).` `223 │ pub fn blocking_rules(firer: LosLevel, target: LosLevel) -> &'static [BlockingRule] {` `224 │     crate::tables_data::LOS_CELLS[firer.index()][target.index()]` `225 │ }`
omdurman-rules/src/los_table.rs:303 GH:omdurman-rules/src/los_table.rs:303	**has_los**	``301 │ /// `unit_level_at` closure returns the LOS level of blocking units`` ``302 │ /// (non-gunboat, non-fort per note a) in an intervening hex, or `None`.`` `303 │ pub fn has_los(` `304 │     board: &crate::board::BoardInfo,` `305 │     from: HexCoord,`

Covered by tests: omdurman-rules::src::effects::tests::can_fire_at_rejects_blocked_los omdurman-rules::src::effects::tests::can_fire_at_allows_clear_los

§6.22 – Consult Range Effects Table

implemented

manual page unknown

> Next consult the Range Effects Table to see if the firing unit's fire combat factor is tripled, doubled, normal, halved, or if the target hex is out of range. Add up the total number of fire combat factors firing at the enemy-occupied hex.

File	Symbol	Code Snippet
omdurman-rules/src/range_effects.rs:34 GH:omdurman-rules/src/range_effects.rs:34	**ae_range_effects**	`/// Look up the range band for an Anglo-Egyptian weapon (§6.22, §6.24).` `/// Distances > 10 are out of range for all weapons.` `pub fn ae_range_effects(weapon: WeaponClass, distance: HexDistance) -> RangeBand {` `    band_at(faction_rows(true), weapon, distance)` `}`
omdurman-rules/src/range_effects.rs:40 GH:omdurman-rules/src/range_effects.rs:40	**dervish_range_effects**	`/// Look up the range band for a Dervish weapon (§6.22).` `/// Distances > 10 are out of range for all weapons.` `pub fn dervish_range_effects(weapon: WeaponClass, distance: HexDistance) -> RangeBand {` `    band_at(faction_rows(false), weapon, distance)` `}`
omdurman-rules/src/lib.rs:519 GH:omdurman-rules/src/lib.rs:519	**RangeBand**	`/// multiplied at a given distance (§6.22).` `#[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)]` `pub enum RangeBand {` `    Tripled,` `    Doubled,`
omdurman-rules/src/lib.rs:187 GH:omdurman-rules/src/lib.rs:187	**HexDistance**	`/// (rulebook §6.22, §7.5).` `#[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Hash, Debug)]` `pub struct HexDistance(u16);` `impl HexDistance {`

Proven by: omdurman-types::src::lib::adjacency_iff_distance_one omdurman-rules::src::lib::fire_factor_row_from_total_matches_printed_bands omdurman-rules::src::lib::range_band_multiplier_arithmetic_is_exact omdurman-rules::src::range_effects::range_effects_are_out_of_range_outside_the_printed_ten_hex_distance_window omdurman-rules::src::range_effects::max_day_range_is_the_last_in_range_hex

Covered by tests: omdurman-rules::src::range_effects::ae_rifles_doubled_at_range_1 omdurman-rules::src::range_effects::ae_rifles_halved_at_range_4 omdurman-rules::src::range_effects::ae_howitzer_range omdurman-rules::src::range_effects::dervish_rifles_shorter_range omdurman-rules::src::range_effects::melee_only_range_1 omdurman-rules::src::range_effects::ae_range_effects_artillery_full omdurman-rules::src::range_effects::ae_range_effects_maxims_match_rifles omdurman-rules::src::range_effects::ae_range_effects_distance_over_10 omdurman-rules::src::range_effects::ae_range_effects_howitzer_halved_4_to_10 omdurman-rules::src::range_effects::dervish_range_effects_rifles omdurman-rules::src::range_effects::dervish_range_effects_artillery omdurman-rules::src::range_effects::dervish_range_effects_maxims_and_howitzer omdurman-rules::src::range_effects::dervish_range_effects_melee omdurman-rules::src::range_effects::dervish_range_effects_distance_over_10 omdurman-rules::src::effects::tests::fire_combat_eliminates_target omdurman-rules::src::range_effects::max_day_range_all_combos omdurman-rules::src::range_effects::range_effects_every_cell_ae_artillery omdurman-rules::src::range_effects::range_effects_every_cell_ae_howitzer omdurman-rules::src::range_effects::range_effects_every_cell_ae_maxims omdurman-rules::src::range_effects::range_effects_every_cell_ae_rifles omdurman-rules::src::range_effects::range_effects_every_cell_dervish_artillery

omdurman-rules::src::range_effects::range_effects_every_cell_dervish_maxims_howitzer
omdurman-rules::src::range_effects::range_effects_every_cell_dervish_rifles
omdurman-rules::src::range_effects::range_effects_every_cell_dervish_spears
omdurman-rules::src::effects::tests::can_fire_at_gates_phase_range_and_player
omdurman-rules::src::effects::tests::mixed_attack_bands_per_firer
omdurman-rules::src::range_effects::ae_range_effects_monotone_non_increasing
omdurman-rules::src::range_effects::ae_howitzer_has_minimum_range
omdurman-rules::src::range_effects::dervish_range_effects_monotone_non_increasing
omdurman-rules::src::range_effects::range_effects_first_range_max_effect_last_range_oor

§6.23 – Terrain defensive modifier

implemented

manual page unknown

> Next check the Terrain Effects Chart to see if the enemy-occupied hex fired upon contains any terrain which gives the enemy units in that hex a defensive benefit. If so, apply this negative modifier to the roll of the ten-sided die and cross-index your net die roll on the Combat Results Table with the number of combat factors firing.

File	Symbol	Code Snippet
omdurman-rules/src/terrain_chart.rs:59 GH:omdurman-rules/src/terrain_chart.rs:59	**defense_modifier**	`/// Convenience: get the defense modifier for a terrain type (rulebook §6.23, Terrain Effects Chart).` `pub fn defense_modifier(terrain: Terrain) -> i16 {` `    terrain_effects_chart(terrain).defense_modifier` `}`
omdurman-rules/src/lib.rs:937 GH:omdurman-rules/src/lib.rs:937	**FireModifier::Terrain**	`    /// Negative modifier from the Terrain Effects Chart applied to the` `    /// defender's hex (§6.23).` `    Terrain(i16),` `    /// -2 thorn-hedge defensive modifier (§9.231).` `    ZaribaThornHedge,`

Proven by: **omdurman-rules::src::lib::fire_modifier_keeps_roll_legal**

Covered by tests: omdurman-rules::src::terrain_chart::clear_terrain_no_bonus omdurman-rules::src::terrain_chart::building_gives_minus_3 omdurman-rules::src::terrain_chart::palm_grove_gives_minus_2 omdurman-rules::src::terrain_chart::rough_movement_and_defense omdurman-rules::src::terrain_chart::swamp_movement_and_defense omdurman-rules::src::terrain_chart::hilltop_movement_and_defense omdurman-rules::src::terrain_chart::huts_movement_and_defense omdurman-rules::src::terrain_chart::defense_modifier_convenience_matches_chart omdurman-rules::src::terrain_chart::terrain_movement_costs_in_bounds omdurman-rules::src::terrain_chart::terrain_defense_modifier_non_positive

§6.24 – Anglo-Egyptian direct fire accuracy bonus and brigade integrity

implemented

manual page unknown

> All Anglo-Egyptian direct fire attacks receive a +1 modifier to their die roll as an accuracy bonus. In addition, any stack of Anglo-Egyptian infantry having brigade integrity (see 5.54) receives a +1 modifier to their die roll if all four fire at the same enemy-occupied hex. These modifiers are cumulative.

See also: §5.54

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:931	**AngloEgyptianDirect-Fire**	`pub enum FireModifier {` `    /// +1 to all Anglo-Egyptian *direct* fire (§6.24).` `    AngloEgyptianDirectFire,`

GH:omdurman-rules/src/lib.rs:931		`932 |     /// +1 brigade integrity, applied only if all four battalions fire at` `933 |     /// the same enemy-occupied hex (§5.54, §6.24).`
omdurman-rules/src/lib.rs:916 GH:omdurman-rules/src/lib.rs:916	BrigadeIntegrity	`914 | /// stack contains all four battalions of a single Anglo-Egyptian brigade.` `915 | #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)]` `916 | pub enum BrigadeIntegrity {` `917 |     None,` `918 |     Integrated(BrigadeId),`
omdurman-rules/src/lib.rs:1031 GH:omdurman-rules/src/lib.rs:1031	FireModifier::die_modifier	`1029 | impl MeleeModifier {` `1030 |     /// Return the numeric die-roll modifier for this melee bonus/penalty (rulebook §7.7, §9.232).` `1031 |     pub fn die_modifier(self) -> i16 {` `1032 |         match self {` `1033 |             MeleeModifier::DervishStandard => 2,`

Proven by: **omdurman-rules::src::lib::die_roll_apply_modifier_is_total**

Covered by tests: **omdurman-rules::src::effects::tests::brigade_integrity_modifier_is_engine_derived** **omdurman-rules::src::effects::tests::fire_modifiers_are_engine_derived_and_mismatches_rejected**

§6.41 – Direct Fire Subphase

implemented

manual page unknown

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:304 GH:omdurman-rules/src/lib.rs:304	DirectFire	`302 | pub enum FireSubPhase {` `303 |     /// Direct fire (§6.41). Both sides participate in this sub-phase.` `304 |     DirectFire,` `305 |     /// Anglo-Egyptian only: Maxim second fire + named-gunboat howitzer fire (§6.42).` `306 |     MaximSecondAndHowitzer,`

Covered by tests: **omdurman-rules::src::effects::tests::disrupted_unit_cannot_fire** **omdurman-rules::src::effects::tests::fire_combat_eliminates_target**

§6.42 – Maxim Second Fire and Howitzer Fire Subphase

implemented

manual page unknown

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:306 GH:omdurman-rules/src/lib.rs:306	MaximSecondAndHowitzer	`304 |     DirectFire,` `305 |     /// Anglo-Egyptian only: Maxim second fire + named-gunboat howitzer fire (§6.42).` `306 |     MaximSecondAndHowitzer,` `307 | }` `308 |`
omdurman-types/src/lib.rs:974 GH:omdurman-types/src/lib.rs:974	fires_twice	`972 |     /// Maxim guns fire twice per turn -- once in the Direct Fire Subphase and` `973 |     /// again in the Maxim Second Fire Subphase (rulebook §6.42).` `974 |     pub fn fires_twice(self) -> bool {` `975 |         matches!(self, UnitKind::Maxim { .. })` `976 |     }`

Covered by tests: **omdurman-rules::src::howitzer_scatter::howitzer_on_target_7_to_10** **omdurman-rules::src::howitzer_scatter::howitzer_scatters_below_7**

`omdurman-rules::src::effects::tests::advance_window_bridges_fire_subphase_and_closes_at_melee`
`omdurman-rules::src::effects::tests::fired_at_tracker_resets_at_maxim_subphase`

§6.51 – Leader Units

implemented

manual page unknown

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:417 GH:omdurman-rules/src/lib.rs:417	**BritishLeader**	`415 │ /// to claim the Mahdi's Tomb (§9.14).` `416 │ #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)]` `417 │ pub enum BritishLeader {` `418 │     Kitchener,` `419 │     Gatacre,`
omdurman-types/src/lib.rs:921 GH:omdurman-types/src/lib.rs:921	**BritishLeader**	`919 │     },` `920 │     /// Anglo-Egyptian leader (§6.51): movement only.` `921 │     BritishLeader { movement: i32 },` `922 │     /// Wall-breach marker placed by artillery fire (§6.63). Not a combat unit.` `923 │     Breech,`
omdurman-types/src/lib.rs:962 GH:omdurman-types/src/lib.rs:962	**has_combat_factors**	`960 │     /// British leaders print a movement factor only (§6.51); other kinds carry` `961 │     /// fire and/or melee factors. Markers carry no stats.` `962 │     pub fn has_combat_factors(self) -> bool {` `963 │         !matches!(` `964 │             self,`

Covered by tests: `omdurman-rules::src::unit_profiles::zero_factor_is_none_not_zero` `omdurman-rules::src::unit_profiles::kitchener_block_resolves_leaders_friendlies_camel_and_sudanese` `omdurman-rules::src::unit_profiles::dervish_leader_sections_resolve_leader_and_retinue_per_cell` `omdurman-rules::src::effects::tests::deploy_rejects_enemy_cohabitation_during_setup` `omdurman-rules::src::effects::tests::dervish_move_through_lone_ae_leader_hex_eliminates_the_leader` `omdurman-rules::src::effects::tests::dervish_move_onto_lone_ae_leader_hex_eliminates_the_leader` `omdurman-rules::src::effects::tests::ae_leader_with_combat_unit_is_not_overrun_by_dervish_move` `omdurman-rules::src::effects::tests::ae_leader_eliminated_with_last_combat_unit_in_fire_combat`

§6.52 – Anglo-Egyptian Friendlies Brigade

implemented

manual page unknown

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:629 GH:omdurman-rules/src/lib.rs:629	**is_friendlies**	`627 │     /// "Friendlies" units obey several special rules (§5.21, §5.23, §6.52,` `628 │     /// §9.14 victory conditions).` `629 │     pub fn is_friendlies(&self) -> bool {` `630 │         matches!(` `631 │             self,`
omdurman-types/src/lib.rs:993 GH:omdurman-types/src/lib.rs:993	**Friendlies**	`991 │     /// Native volunteer brigade -- the Shaggyeh (§6.52). Do not receive` `992 │     /// brigade integrity (§5.54 enumerates only British/Egyptian/Sudanese).` `993 │     Friendlies,` `994 │ }` `995 │`

Covered by tests: `omdurman-rules::src::unit_profiles::friendlies_counters_score_by_bank_not_as_leaders` `omdurman-rules::src::effects::tests::friendlies_validate_and_resolve_on_dervish_table`

§6.53 – Royal Engineers demolition

implemented

manual page unknown

<table>
<tr><th>File</th><th>Symbol</th><th>Code Snippet</th></tr>
<tr><td>omdurman-rules/src/lib.rs:600
GH:omdurman-rules/src/lib.rs:600</td><td>RoyalEngineers</td><td><pre>
598 | /// The Royal Engineers (§6.53) -- a *specific* unit, not a class, so we
599 | /// model it explicitly.
600 | RoyalEngineers,
601 | }
602 |
</pre></td></tr>
<tr><td>omdurman-rules/src/lib.rs:835
GH:omdurman-rules/src/lib.rs:835</td><td>demolishing</td><td><pre>
833 | /// Set when the Royal Engineers are committed to a demolition this turn
834 | /// (§6.53) -- neither offensive fire nor melee allowed that turn.
835 | pub demolishing: bool,
836 | /// Set when a gunboat has lost its engines to a river mine (§10.12, roll
837 | /// 5-7): it may no longer move under power and instead drifts two hexes per
</pre></td></tr>
<tr><td>omdurman-rules/src/effects/effect.rs:175
GH:omdurman-rules/src/effects/effect.rs:175</td><td>Demolition</td><td><pre>
173 |
174 | /// Royal Engineers demolition (rulebook §6.53).
175 | Demolition {
176 | unit_id: UnitId,
177 | target: DemolitionTarget,
</pre></td></tr>
<tr><td>omdurman-rules/src/effects/setup.rs:35
GH:omdurman-rules/src/effects/setup.rs:35</td><td>apply_demolition</td><td><pre>
 33 | /// resolution happens at end of turn via [`apply_resolve_demolition`], which
 34 | /// checks the engineer is still adjacent and undisrupted.
 35 | pub fn apply_demolition(
 36 | state: &mut GameState,
 37 | unit_id: UnitId,
</pre></td></tr>
<tr><td>omdurman-rules/src/lib.rs:1061
GH:omdurman-rules/src/lib.rs:1061</td><td>DemolitionTarget</td><td><pre>
1059 | /// disrupted or driven off.
1060 | #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)]
1061 | pub enum DemolitionTarget {
1062 | Fort(UnitId),
1063 | WallHexside(HexsideRef),
</pre></td></tr>
<tr><td>omdurman-rules/src/effects/state.rs:2015
GH:omdurman-rules/src/effects/state.rs:2015</td><td>demolition_targets</td><td><pre>
2013 | /// the rules would accept. Empty when the unit doesn't exist or has no
2014 | /// adjacent target.
2015 | pub fn demolition_targets(&self, unit_id: UnitId) -> Vec<DemolitionTarget> {
2016 | let Ok(unit) = self.unit_or_err(unit_id) else {
2017 | return Vec::new();
</pre></td></tr>
</table>

Covered by tests: omdurman-rules::src::effects::tests::demolition_cancelled_when_engineer_disrupted omdurman-rules::src::effects::tests::demolition_cancelled_when_engineer_moved_away omdurman-rules::src::effects::tests::demolition_targets_finds_adjacent_fort_and_wall

§6.54 – Forts

implemented

manual page unknown

<table>
<tr><th>File</th><th>Symbol</th><th>Code Snippet</th></tr>
<tr><td>omdurman-rules/src/lib.rs:868</td><td>ZocReason</td><td><pre>
866 | /// Used by the engine when answering "is this hex in an enemy ZOC?".
867 | #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)]
</pre></td></tr>
</table>

GH:omdurman-rules/src/lib.rs:868		`868 │ pub enum ZocReason {` `869 │     /// Normal ZOC: any non-disrupted unit other than an Anglo-Egyptian` `870 │     /// leader (§5.41) projects ZOC into each of its six adjacent hexes.`
omdurman-rules/src/lib.rs:875 GH:omdurman-rules/src/lib.rs:875	**Fort**	`873 │     GunboatVsGunboat,` `874 │     /// Forts project ZOC out of, but not into, an empty fort (§5.44, §6.54).` `875 │     Fort,` `876 │     /// Walled-city ZOC: extends out through walls and gates but not in,` `877 │     /// across a breach in both directions (§5.44).`
omdurman-rules/src/lib.rs:823 GH:omdurman-rules/src/lib.rs:823	**UnitState**	`821 │ /// rather than one big enum.` `822 │ #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug, Default)]` `823 │ pub struct UnitState {` `824 │     /// Reference table: "Disrupted units: no ZOC; may not move; may not fire` `825 │     /// offensively or defensively; may not melee; are turned face up at the`
omdurman-rules/src/lib.rs:976 GH:omdurman-rules/src/lib.rs:976	**FireAttack**	`974 │ /// modifiers (rulebook §6).` `975 │ #[derive(Serialize, Deserialize, Clone, Debug, PartialEq, Eq)]` `976 │ pub struct FireAttack {` `977 │     pub firing_player: Player,` `978 │     pub phase: Phase,`
omdurman-rules/src/lib.rs:991 GH:omdurman-rules/src/lib.rs:991	**FireAttack::net_modifier**	`989 │ impl FireAttack {` `990 │     /// Sum of all fire modifiers applied to this attack (rulebook §6.24).` `991 │     pub fn net_modifier(&self) -> i16 {` `992 │ // Saturating fold rather than ``sum()``: the modifier list is unbounded` `993 │ // and arrives over the network, so a plain sum can overflow ``i16``.`

Covered by tests: **omdurman-rules::src::effects::tests::retreat_before_melee_may_not_land_on_enemy_fort**

§6.61 – Only artillery may fire at gunboats; 3+ to sink

implemented

manual page unknown

> Only artillery may fire at gunboats. A result of 3 or more on the combat results table is required to sink a gunboat. Any other result is a miss.

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:476 GH:omdurman-rules/src/lib.rs:476	**WeaponClass**	`474 │     Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Hash, Debug, strum::Display,` `475 │ )]` `476 │ pub enum WeaponClass {` `477 │     /// Dervish spears and swords -- no ranged fire at all.` `478 │     Melee,`

Covered by tests: **omdurman-rules::src::effects::tests::rifles_may_not_sink_a_gunboat** **omdurman-rules::src::effects::tests::artillery_sinks_gunboat_only_on_three_plus**

§6.62 – Only artillery may fire at forts; 2+ to destroy

implemented

manual page unknown

> Only artillery may fire at forts. A result of 2 or more on the combat results table is required to eliminate a fort. Any other result is a miss. If the fort contains any enemy units at the instant it is destroyed, one unit is eliminated with the fort.

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:476 GH:omdurman-rules/src/lib.rs:476	**WeaponClass**	`474 │     Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Hash, Debug, strum::Display,` `475 │ )]` `476 │ pub enum WeaponClass {` `477 │     /// Dervish spears and swords -- no ranged fire at all.` `478 │     Melee,`

Covered by tests: **omdurman-rules::src::effects::tests::artillery_destroys_fort_on_two_or_better_only**

§6.63 – Only artillery may breach wall hexsides; 2+ required

implemented

manual page unknown

> Only artillery may fire to breach a wall hexside of Khartoum or the walled city of Omdurman. A result of 2 or more on the combat results table is required to breach a wall. Any other result is a miss. The effect of the breach is to negate the wall hexside for line of sight purposes. Place a “BREACH” marker in an adjacent hex so that the arrow points to the breached hexside. If any enemy units are adjacent to the wall hexside at the instant it is breached, one enemy unit is eliminated.
>
> **6.64) Howitzer fire:** Five units in the game have howitzer fire capability. These are the five named British gunboats. They may fire their artillery factor as direct fire during the Direct Fire Subphase (see 4 and 6.41) and may then fire the same artillery factor as howitzer fire during the Maxim Second Fire and Howitzer Subphase (see 4 and 6.42). Exception: no howitzer fire is allowed during night game turns. To fire howitzer fire, select any target hex between 4 and 10 hexes from the firing gunboat (ignoring the Line of Sight Table) and roll the ten-sided die twice. The first die roll is the Combat Results Table die roll and the second roll is the impact hex die roll. Refer to the Howitzer Fire Scattergram on the mapsheet for the impact hex. The designated target hex is hit on a roll of 7–10. Once a howitzer

(see manual for full text)

See also: §6.41, §6.42, §6.64

File	Symbol	Code Snippet
omdurman-types/src/lib.rs:198 GH:omdurman-types/src/lib.rs:198	**Breach**	`196 │     /// Breach in a wall (artillery/§6.63 or Royal Engineers/§6.53). ZOC both` `197 │     /// ways; LOS no longer blocked across the hexside.` `198 │     Breach,` `199 │     /// Khor -- gully/wadi. ZOCs do not extend across (§5.44); advance after` `200 │     /// combat may not cross (§6.82).`
omdurman-rules/src/effects/effect.rs:278 GH:omdurman-rules/src/effects/effect.rs:278	**ArtilleryBreachWall**	`276 │     /// pre-rolled d10 used for the CRT lookup; range/LOS are re-derived by the` `277 │     /// engine from the firers and `target`.` `278 │     ArtilleryBreachWall {` `279 │         firers: Vec<UnitId>,` `280 │         target: HexsideRef,`
omdurman-rules/src/effects/fire.rs:607 GH:omdurman-rules/src/effects/fire.rs:607	**apply_artillery_breach_w**	`605 │ /// artillery's CRT roll -- the rulebook specifies the same "2+ required"` `606 │ /// threshold for both trigger styles.` `607 │ pub fn apply_artillery_breach_wall(` `608 │     state: &mut GameState,` `609 │     firers: &[UnitId],`
omdurman-rules/src/effects/state.rs:1215 GH:omdurman-	**can_fire_at_wall**	`1213 │     /// range band and resolving the CRT — this method only validates one` `1214 │     /// firer at a time.` `1215 │     pub fn can_fire_at_wall(`

rules/src/effects/ state.rs:1215		`1216 │         &self,` `1217 │         firer: UnitId,`

Covered by tests: **omdurman-rules::src::unit_profiles::breech_marker_cell_returns_none** **omdurman-rules::src::effects::tests::artillery_breaches_wall_only_on_crt_two_or_better** **omdurman-rules::src::effects::tests::wall_breach_eliminates_one_adjacent_enemy_unit**

§6.64 – Howitzer fire

implemented

manual page unknown

File	Symbol	Code Snippet
omdurman-rules/src/howitzer_scatter.rs:8 GH:omdurman-rules/src/howitzer_scatter.rs:8	**ScatterHexDirection**	`6 │ /// the diagram away from the firing player.` `7 │ #[derive(serde::Serialize, serde::Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug)]` `8 │ pub enum ScatterHexDirection {` `9 │     UpperLeft,` `10 │     UpperRight,`
omdurman-rules/src/howitzer_scatter.rs:38 GH:omdurman-rules/src/howitzer_scatter.rs:38	**howitzer_scatter**	`36 │ /// onto a hex-grid offset oriented away from the firer (see` `37 │ /// ``GameState::howitzer_impact_hex``).` `38 │ pub fn howitzer_scatter(impact_roll: DieRoll) -> ScatterHexDirection {` `39 │ crate::tables_data::SCATTERGRAM[(impact_roll.value() - 1) as usize]` `40 │ }`
omdurman-rules/src/lib.rs:428 GH:omdurman-rules/src/lib.rs:428	**GunboatId**	`426 │ /// fire; "old" gunboats do not (rulebook §2.32).` `427 │ #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)]` `428 │ pub enum GunboatId {` `429 │     /// One of the five new-type named gunboats with howitzer capability.` `430 │     Named(NamedGunboat),`
omdurman-rules/src/effects/effect.rs:88 GH:omdurman-rules/src/effects/effect.rs:88	**HowitzerFire**	`86 │     /// - CRT result applied to units at impact hex (not the original target).` `87 │     /// - Firers marked as fired.` `88 │     HowitzerFire {` `89 │         attack: FireAttack,` `90 │         combat_results_table_roll: DieRoll,`
omdurman-rules/src/effects/fire.rs:13 GH:omdurman-rules/src/effects/fire.rs:13	**apply_howitzer_fire**	`11 │` `12 │ /// Validate and apply a howitzer fire attack (scatter path) (rulebook §6.64).` `13 │ pub fn apply_howitzer_fire(` `14 │     state: &mut GameState,` `15 │     attack: &FireAttack,`
omdurman-rules/src/effects/state.rs:1030 GH:omdurman-rules/src/effects/state.rs:1030	**can_fire_at**	`1028 │ /// modifier in the [``FireAttack``] and is responsible for the LOS gate.` `1029 │ /// (Howitzer fire ignores LOS entirely -- §6.64.)` `1030 │ pub fn can_fire_at(` `1031 │ &self,` `1032 │ firer: UnitId,`

Proven by: **omdurman-rules::src::effects::opposite_is_an_involution** **omdurman-rules::src::effects::step_toward_lands_on_an_adjacent_hex** **omdurman-rules::src::howitzer_scatter::scatter_is_center_exactly_for_rolls_7_to_10**

Covered by tests: **omdurman-rules::src::howitzer_scatter::howitzer_on_target_7_to_10** **omdurman-rules::src::howitzer_scatter::howitzer_scatters_below_7** **omdurman-rules::src::howitzer_scatter::howitzer_each_miss_gets_its_printed_hex** **omdurman-rules::src::unit_profiles::named_and_old_gunboats_resolve**

omdurman-rules::src::effects::tests::named_gunboat_has_howitzer
omdurman-rules::src::effects::tests::named_gunboat_may_fire_howitzer_in_second_subphase
omdurman-rules::src::effects::tests::named_gunboat_direct_fire_uses_artillery_weapon
omdurman-rules::src::effects::tests::named_gunboat_no_howitzer_at_night
omdurman-rules::src::effects::tests::dervish_gunboat_lacks_howitzer

§6.81 – Moving player may fire with all capable units

implicit

manual page unknown

> During Offensive Fire phase, the moving player may fire with all of his units capable of firing, up to their maximum range, within the limitations imposed by the rules of combat.

§6.82 – Advance after combat (offensive fire)

implemented

manual page unknown

> If an enemy-occupied hex is vacated as a result of offensive fire, friendly units may advance after combat into the vacated hex. To be eligible to advance, the friendly units must have participated in the attack and must have been adjacent to the vacated hex. Note that artillery may not advance, nor may units advance across a wall hexside (except at a gate or breach). Units may never advance after combat across a khor.

<table>
<tr><th>File</th><th>Symbol</th><th>Code Snippet</th></tr>
<tr><td>omdurman-rules/src/effects/effect.rs:162
GH:omdurman-rules/src/effects/effect.rs:162</td><td>AdvanceAfterCombat</td><td>160 | /// **Postconditions:** Unit position moved to `to`; `vacated_by_combat`
161 | /// entry consumed.
162 | AdvanceAfterCombat { unit_id: UnitId, to: HexCoord },
163 |
164 | // -- Unit state changes
--</td></tr>
<tr><td>omdurman-rules/src/effects/dispatch.rs:36
GH:omdurman-rules/src/effects/dispatch.rs:36</td><td>apply_advance_after_cor</td><td>34 | }
35 | GameEffect::AdvanceAfterCombat { unit_id, to } => {
36 | apply_advance_after_combat(state, *unit_id, *to)
37 | }
38 | GameEffect::RecoverUnit { unit_id } => apply_recover_unit(state, *unit_id),</td></tr>
<tr><td>omdurman-types/src/lib.rs:241
GH:omdurman-types/src/lib.rs:241</td><td>blocks_advance_after_co</td><td>239 |
240 | /// Whether advance-after-combat may *not* cross this side (§6.82, §7.6).
241 | pub fn blocks_advance_after_combat(self) -> bool {
242 | matches!(
243 | self,</td></tr>
<tr><td>omdurman-rules/src/effects/state.rs:1919
GH:omdurman-rules/src/effects/state.rs:1919</td><td>can_advance_after_comb</td><td>1917 | /// player's unit, not artillery, adjacent to `to`, and `to` now empty.
1918 | /// Wall/khor hexside restrictions are not enforced (no hexside map data).
1919 | pub fn can_advance_after_combat(&self, unit_id: UnitId, to: HexCoord) -> Result<(), RuleError> {
1920 | let unit = self.unit_or_err(unit_id)?;
1921 | // §6.7: there is no advance after combat as a result of defensive fire.</td></tr>
</table>

Proven by: omdurman-rules::src::effects::advance_phase_is_atomic

Covered by tests: omdurman-rules::src::effects::tests::can_advance_after_combat_rejects_wall_hexside
omdurman-rules::src::effects::tests::can_advance_after_combat_rejects_khor_hexside
omdurman-rules::src::effects::tests::advance_requires_combat_vacated_hex

omdurman-rules::src::effects::tests::advance_requires_participation

omdurman-rules::src::effects::tests::rejected_advance_phase_keeps_vacated_windows

8/8 implemented

§7 – Melee Phase

§7 – Melee Phase (chapter)

implemented

manual page unknown

> Melee Phase

File	Symbol	Code Snippet
omdurman-rules/src/effects/state.rs:133 GH:omdurman-rules/src/effects/state.rs:133	PendingMelee	`/// resolution after the reaction window is deterministic and host-ordered (rulebook §7.5).` `#[derive(Serialize, Deserialize, Clone, Debug)]` `pub struct PendingMelee {` `    pub attack: MeleeAttack,` `    pub attacker_roll: DieRoll,`

Covered by tests: **omdurman-rules::src::effects::tests::declared_melee_blocks_phase_advance**

§7.1 – Melee strength printed on counter

implemented

manual page unknown

> The melee strength of all units is printed on the counter. Note that gunboats have no melee strength. Gunboats may neither melee attack nor be melee attacked.

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:110 GH:omdurman-rules/src/lib.rs:110	MeleeFactor	`    /// Every possible value from the annotated counter set is a named variant.` `    #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)]` `    pub enum MeleeFactor {` `        One = 1,` `        Three = 3,`
omdurman-rules/src/lib.rs:121 GH:omdurman-rules/src/lib.rs:121	MeleeFactor::sum	`impl MeleeFactor {` `    /// Sum multiple melee factors (rulebook §7.1).` `    pub fn sum<'a>(factors: impl IntoIterator<Item = &'a MeleeFactor>) -> u16 {` `        factors.into_iter().map(|f| f.value()).sum()` `    }`
omdurman-types/src/lib.rs:945 GH:omdurman-types/src/lib.rs:945	may_be_melee_attacked	 `    /// Gunboats neither attack nor are attacked in melee (§7.1).` `    pub fn may_be_melee_attacked(self) -> bool {` `        !matches!(self, UnitKind::Gunboat { .. })` `    }`

Covered by tests: **omdurman-rules::src::lib::melee_factor_values_and_sum** **omdurman-rules::src::lib::unit_kind_melee_capability**

§7.2 – Melee adjacent only, not across wall hexsides

implemented

manual page unknown

> Melee simulates the hand-to-hand fighting of the period. Units may melee attack adjacent enemy units only. Units may not melee attack across a wall hexside, but may melee attack through a gate or breach hexside.

File	Symbol	Code Snippet
omdurman-types/src/lib.rs:236 GH:omdurman-types/src/lib.rs:236	blocks_melee	`    /// Whether melee may *not* be made across this side (§7.2). Gates and` `    /// breaches are passable to melee.` `    pub fn blocks_melee(self) -> bool {` `        matches!(self, HexsideKind::Wall | HexsideKind::ZaribaThornHedge)` `    }`
omdurman-rules/src/effects/state.rs:1305 GH:omdurman-rules/src/effects/state.rs:1305	can_melee	`    /// that may be melee-attacked (gunboats may not -- §7.1), and no wall or` `    /// thorn-hedge hexside blocks the attack (§7.2).` `    pub fn can_melee(&self, attacker: UnitId, defender_hex: HexCoord) -> Result<(), RuleError> {` `        let unit = self.unit_or_err(attacker)?;`

Covered by tests: omdurman-rules::src::effects::tests::can_melee_gates_phase_adjacency_and_kind omdurman-rules::src::effects::tests::can_melee_rejects_wall_hexside omdurman-rules::src::effects::tests::can_melee_rejects_thorn_hedge_hexside omdurman-rules::src::effects::tests::can_melee_allows_gate_hexside

§7.3 – Simultaneous melee combat

implemented

manual page unknown

> Melee combat is considered simultaneous, so that units eliminated by melee attacks still get a melee combat die roll.

File	Symbol	Code Snippet
omdurman-rules/src/effects/effect.rs:111 GH:omdurman-rules/src/effects/effect.rs:111	MeleeCombat	`    /// - Winner may advance into vacated hex (§7.6).` `    /// - Victory points awarded for eliminations.` `    MeleeCombat {` `        attack: MeleeAttack,` `        attacker_roll: DieRoll,`
omdurman-rules/src/effects/melee.rs:4 GH:omdurman-rules/src/effects/melee.rs:4	apply_melee_combat	`/// Apply a simultaneous melee combat between two adjacent hexes (rulebook §7).` `pub fn apply_melee_combat(` `    state: &mut GameState,` `    attack: &MeleeAttack,`

Covered by tests: omdurman-rules::src::effects::tests::melee_resolves_simultaneously

§7.4 – Who may melee attack / defend

implemented

manual page unknown

> Only infantry, cavalry, camel units, and Dervish leaders may melee attack. All units (except gunboats — see 7.1) may melee defend.

See also: §7.1

File	Symbol	Code Snippet
omdurman-types/src/lib.rs:934	may_melee_attack	`    /// Rulebook §7.4 -- only infantry, cavalry, camel and Dervish leaders may` `    /// melee *attack*. All others (except gunboats) may melee *defend* (§7.1).`

<table>
<tr><td>GH:omdurman-types/src/lib.rs:934</td><td></td><td>934 | pub fn may_melee_attack(self) -> bool {
935 | matches!(
936 | self,</td></tr>
<tr><td>omdurman-types/src/lib.rs:871
GH:omdurman-types/src/lib.rs:871</td><td>UnitKind</td><td>869 | /// `Some(UnitKind::Marker)` or `None`.
870 | #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)]
871 | pub enum UnitKind {
872 | /// Foot infantry (§2.3): fire / melee / movement.
873 | Infantry {</td></tr>
<tr><td>omdurman-types/src/lib.rs:713
GH:omdurman-types/src/lib.rs:713</td><td>DervishTribe</td><td>711 | Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display, strum::EnumIter,
712 |)]
713 | pub enum DervishTribe {
714 | Baggara,
715 | Jaalin,</td></tr>
<tr><td>omdurman-rules/src/effects/state.rs:1305
GH:omdurman-rules/src/effects/state.rs:1305</td><td>can_melee</td><td>1303 | /// that may be melee-attacked (gunboats may not -- §7.1), and no wall or
1304 | /// thorn-hedge hexside blocks the attack (§7.2).
1305 | pub fn can_melee(&self, attacker: UnitId, defender_hex: HexCoord) -> Result<(), RuleError> {
1306 | let unit = self.unit_or_err(attacker)?;
1307 |</td></tr>
</table>

Covered by tests: **omdurman-rules::src::effects::tests::can_melee_gates_phase_adjacency_and_kind**

§7.5 – Cavalry/camel retreat before melee

implemented

manual page unknown

> Cavalry and camel units may retreat two hexes from an infantry melee attack. Note, however, that only one retreat per unit per turn is permitted. Thus, if their retreat places them adjacent to enemy units whose melee attacks have not yet been resolved, those enemy units may elect to attack the retreating unit(s).

<table>
<tr><th>File</th><th>Symbol</th><th>Code Snippet</th></tr>
<tr><td>omdurman-rules/src/effects/effect.rs:148
GH:omdurman-rules/src/effects/effect.rs:148</td><td>RetreatBeforeMelee</td><td>146 | ///
147 | /// **Postconditions:** Unit position moved to `to`.
148 | RetreatBeforeMelee { unit_id: UnitId, to: HexCoord },
149 |
150 | /// An attacking unit advances into a hex vacated by combat (rulebook §6.82</td></tr>
<tr><td>omdurman-rules/src/effects/dispatch.rs:33
GH:omdurman-rules/src/effects/dispatch.rs:33</td><td>apply_retreat_before_me</td><td> 31 | GameEffect::ResolveMelee => apply_resolve_melee(state),
 32 | GameEffect::RetreatBeforeMelee { unit_id, to } => {
 33 | apply_retreat_before_melee(state, *unit_id, *to)
 34 | }
 35 | GameEffect::AdvanceAfterCombat { unit_id, to } => {</td></tr>
<tr><td>omdurman-types/src/lib.rs:951
GH:omdurman-types/src/lib.rs:951</td><td>may_retreat_before_mel</td><td>949 | /// Cavalry and camel units may retreat two hexes from an infantry melee
950 | /// attack (§7.5).
951 | pub fn may_retreat_before_melee(self) -> bool {
952 | matches!(self, UnitKind::Cavalry { .. } | UnitKind::Camel { .. })
953 | }</td></tr>
<tr><td>omdurman-rules/src/lib.rs:187</td><td>HexDistance</td><td>185 | /// (rulebook §6.22, §7.5).
186 | #[derive(Serialize, Deserialize, Clone, Copy,</td></tr>
</table>

GH:omdurman-rules/src/lib.rs:187		`PartialEq, Eq, PartialOrd, Ord, Hash, Debug)]` `pub struct HexDistance(u16);` `impl HexDistance {`
omdurman-rules/src/effects/state.rs:1850 GH:omdurman-rules/src/effects/state.rs:1850	**can_retreat_before_mele**	`    /// two hexes away and empty. (Does not verify the attacker is infantry --` `    /// the caller offers the retreat only in response to one.)` `    pub fn can_retreat_before_melee(&self, unit_id: UnitId, to: HexCoord) -> Result<(), RuleError> {` `        let unit = self.unit_or_err(unit_id)?;` `        if !matches!(self.phase, Phase::Melee) {`

Proven by: **omdurman-rules::src::effects::resolve_melee_is_atomic** **omdurman-rules::src::effects::advance_phase_is_atomic**

Covered by tests: **omdurman-rules::src::effects::tests::retreat_before_melee_only_cavalry_two_hexes** **omdurman-rules::src::effects::tests::retreat_opens_window_only_when_hex_empties** **omdurman-rules::src::effects::tests::rejected_resolve_melee_keeps_the_declaration**

§7.6 – Advance after melee

implemented

manual page unknown

> If a melee attack eliminates all of the defenders in an adjacent hex, the Dervish player MUST advance into the vacated hex. To be eligible to advance, the Dervish units must have been adjacent to the vacated hex and participated in the melee attack that eliminated the defenders. All surviving eligible Dervish units MUST advance, up to the stacking limit. The Anglo-Egyptian player may advance if desired. Note that only attacking units may advance.

File	Symbol	Code Snippet
omdurman-rules/src/effects/effect.rs:162 GH:omdurman-rules/src/effects/effect.rs:162	**AdvanceAfterCombat**	`    /// **Postconditions:** Unit position moved to `to`; `vacated_by_combat`` `    /// entry consumed.` `    AdvanceAfterCombat { unit_id: UnitId, to: HexCoord },` `    // -- Unit state changes` `------------------------------------------------`
omdurman-rules/src/effects/dispatch.rs:36 GH:omdurman-rules/src/effects/dispatch.rs:36	**apply_advance_after_com**	`        }` `        GameEffect::AdvanceAfterCombat { unit_id, to } => {` `            apply_advance_after_combat(state, *unit_id, *to)` `        }` `        GameEffect::RecoverUnit { unit_id } => apply_recover_unit(state, *unit_id),`
omdurman-rules/src/effects/state.rs:1919 GH:omdurman-rules/src/effects/state.rs:1919	**can_advance_after_comb**	`    /// player's unit, not artillery, adjacent to `to`, and `to` now empty.` `    /// Wall/khor hexside restrictions are not enforced (no hexside map data).` `    pub fn can_advance_after_combat(&self, unit_id: UnitId, to: HexCoord) -> Result<(), RuleError> {` `        let unit = self.unit_or_err(unit_id)?;` `        // §6.7: there is no advance after combat as a result of defensive fire.`

Proven by: **omdurman-rules::src::effects::advance_phase_is_atomic**

Covered by tests: **omdurman-rules::src::effects::tests::dervish_advance_after_melee_is_mandatory** **omdurman-rules::src::effects::tests::rejected_advance_phase_keeps_vacated_windows** **omdurman-rules::src::effects::tests::melee_mandatory_advance_leaders_are_free_stacking**

§7.7 – Melee modifiers

implemented

manual page unknown

> To resolve melee, both the attacker and the defender roll on the Combat Results Table and apply the applicable melee modifier to their die roll. The Dervish player receives a +2 melee modifier, the Anglo-Egyptian player receives a +1 melee modifier. No terrain modifiers are applied to melee combat (Exception: Zariba hexsides in the historical scenario and the campaign game, if constructed — see 9.23). Melee losses must be taken from meleeing units first!

See also: §9.23

<table>
<tr><th>File</th><th>Symbol</th><th>Code Snippet</th></tr>
<tr><td>omdurman-rules/src/lib.rs:1019
GH:omdurman-rules/src/lib.rs:1019</td><td>MeleeModifier</td><td><pre>1017 |
1018 | #[derive(Serialize, Deserialize, Clone, Copy,
PartialEq, Eq, Debug)]
1019 | pub enum MeleeModifier {
1020 | /// +2 to all Dervish melee rolls (§7.7).
1021 | DervishStandard,</pre></td></tr>
<tr><td>omdurman-rules/src/lib.rs:1042
GH:omdurman-rules/src/lib.rs:1042</td><td>MeleeAttack</td><td><pre>1040 | /// A melee attack: simultaneous, both sides roll on
the Combat Results Table (§7.3, §7.7).
1041 | #[derive(Serialize, Deserialize, Clone, Debug,
PartialEq, Eq)]
1042 | pub struct MeleeAttack {
1043 | pub attacker_player: Player,
1044 | pub attacker_hex: HexCoord,</pre></td></tr>
<tr><td>omdurman-rules/src/lib.rs:1023
GH:omdurman-rules/src/lib.rs:1023</td><td>MeleeModifier::AngloEgyptianStandard</td><td><pre>1021 | DervishStandard,
1022 | /// +1 to all Anglo-Egyptian melee rolls (§7.7).
1023 | AngloEgyptianStandard,
1024 | /// Inverted to -2 when Dervish units melee-attack
across a trench into
1025 | /// an entrenched defender (§9.232).</pre></td></tr>
<tr><td>omdurman-rules/src/lib.rs:1026
GH:omdurman-rules/src/lib.rs:1026</td><td>MeleeModifier::DervishV Defender</td><td><pre>1024 | /// Inverted to -2 when Dervish units melee-attack
across a trench into
1025 | /// an entrenched defender (§9.232).
1026 | DervishVsTrenchedDefender,
1027 | }
1028 |</pre></td></tr>
<tr><td>omdurman-rules/src/lib.rs:1021
GH:omdurman-rules/src/lib.rs:1021</td><td>MeleeModifier::DervishStandard</td><td><pre>1019 | pub enum MeleeModifier {
1020 | /// +2 to all Dervish melee rolls (§7.7).
1021 | DervishStandard,
1022 | /// +1 to all Anglo-Egyptian melee rolls (§7.7).
1023 | AngloEgyptianStandard,</pre></td></tr>
</table>

Proven by: **omdurman-rules::src::lib::melee_modifier_keeps_roll_legal**

Covered by tests: **omdurman-rules::src::effects::tests::melee_resolves_simultaneously** **omdurman-rules::src::effects::tests::melee_modifiers_are_engine_derived_and_mismatches_rejected**

2/3 implemented

§8 – Night Game Turns

§8 – Night Game Turns

descriptive

manual page unknown

> Night Game Turns

§8.1 – Night effects

implemented

manual page unknown

> The effects of night game turns are: a) all Anglo-Egyptian movement allowances are halved (round down), b) there is no Anglo-Egyptian howitzer fire, and c) all fire ranges are halved for both sides (round down, but range 1 stays range 1). Range effects on fire combat are the same as during day game turns. For example, an Anglo-Egyptian infantry unit firing at night will be doubled at range 1, normal at range 2, and may not fire at range 3 or greater.
>
> **8.2) Dervish Desertion Roll:** Once each campaign game, during the first night turn of the game, the Dervish player rolls one die to see how many of his units desert. The roll is made during the movement phase and the number of deserting Dervish units is equal to 1½ times the roll of one die. The Dervish player may choose which units desert by merely removing them from the mapsheet. The KHALIFA unit, gunboats, artillery units, and forts are the only Dervish units that may not be chosen. No victory points are awarded to the Anglo-Egyptian player for deserting Dervishes.

(see manual for full text)

See also: §8.2

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:156 GH:omdurman-rules/src/lib.rs:156	**MovementAllowance::halve**	`154 │ impl MovementAllowance {` `155 │     /// Night movement allowance = halved (round down) (rulebook §8.1, §5.11).` `156 │     pub fn halve(self) -> Self {` `157 │         let v = self.value() / 2;` `158 │         MovementAllowance::try_from(v).expect("halved value always a named variant")`
omdurman-types/src/lib.rs:778 GH:omdurman-types/src/lib.rs:778	**DayNight**	`776 │ /// (rulebook §8.1).` `777 │ #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)]` `778 │ pub enum DayNight {` `779 │     Day,` `780 │     Night,`
omdurman-rules/src/range_effects.rs:71 GH:omdurman-rules/src/range_effects.rs:71	**night_max_range**	`69 │` `70 │ /// The halved maximum range at night (§8.1): round down, minimum 1.` `71 │ pub fn night_max_range(weapon: WeaponClass, ae: bool) -> u8 {` `72 │     let day = max_day_range(weapon, ae);` `73 │     if day <= 1 { 1 } else { day / 2 }`
omdurman-rules/src/lib.rs:1514 GH:omdurman-rules/src/lib.rs:1514	**effective_movement_at_**	`1512 │ /// Apply night-turn movement halving for Anglo-Egyptian units (§8.1): all` `1513 │ /// Anglo-Egyptian movement allowances are halved (round down).` `1514 │ pub fn effective_movement_at_night(` `1515 │     allowance: MovementAllowance,` `1516 │     player: Player,`

Proven by: omdurman-rules::src::lib::movement_allowance_halve_never_panics omdurman-rules::src::lib::night_halving_is_ae_only_and_day_neutral omdurman-rules::src::range_effects::night_cap_is_halved_day_max_floored_at_one omdurman-rules::src::range_effects::night_range_effects_gates_the_day_band_by_the_cap

Covered by tests: omdurman-rules::src::range_effects::night_max_ranges omdurman-rules::src::range_effects::night_max_ranges_remaining omdurman-rules::src::range_effects::ae_rifle_at_night_matches_rulebook_example omdurman-rules::src::range_effects::max_day_range_all_combos omdurman-rules::src::effects::tests::night_movement_overlay_allowance_halved

§8.2 – Dervish Desertion Roll

implemented

manual page unknown

File	Symbol	Code Snippet
omdurman-rules/src/effects/effect.rs:190 GH:omdurman-rules/src/effects/effect.rs:190	DervishDesertion	`188 │     /// the effect. The Khalifa, gunboats, artillery, and forts may not be` `189 │     /// chosen.` `190 │     DervishDesertion {` `191 │         roll: DieRoll,` `192 │         deserters: Vec<UnitId>,`
omdurman-rules/src/turn_track.rs:63 GH:omdurman-rules/src/turn_track.rs:63	DervishDesertion	`61 │     None,` `62 │     /// Dervish desertion roll (§8.2) -- occurs on the first night turn.` `63 │     DervishDesertion,` `64 │     /// Dervish reinforcements are available.` `65 │     DervishReinforcements,`
omdurman-rules/src/turn_track.rs:60 GH:omdurman-rules/src/turn_track.rs:60	TurnEvent	`58 │ /// Special events that occur on specific turns (rulebook §8.2, §9.112, §9.113).` `59 │ #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)]` `60 │ pub enum TurnEvent {` `61 │     None,` `62 │     /// Dervish desertion roll (§8.2) -- occurs on the first night turn.`

Proven by: omdurman-rules::src::effects::desertion_count_is_one_and_a_half_times_the_roll

Covered by tests: omdurman-rules::src::effects::tests::desertion_count_is_floor_one_and_a_half omdurman-rules::src::turn_track::desertion_on_first_night omdurman-rules::src::effects::tests::desertion_roll_required_before_first_night_movement_ends

23/33 implemented

§9 – The Scenarios

§9 – The Scenarios

descriptive

manual page unknown

> The Scenarios

§9.1 – The Campaign Game

descriptive

manual page unknown

> The Campaign Game

§9.2 – The Historical Scenario

descriptive

manual page unknown

> The Historical Scenario
>
> Players should note that the historical scenario is an exercise in futility for the Dervish player. It is, however, an interesting demonstration of the absolute imbecility of the Khalifa's generalship and vividly shows the superiority of entrenched firepower over traditional tribal arms in the colonial period.

§9.3 – Bonus Game: Fall of Khartoum

descriptive

manual page unknown

> Bonus Game: FALL OF KHARTOUM

§9.11 – Set Up (Campaign)

descriptive

manual page unknown

> Set Up

§9.12 – Scenario Length (Campaign)

implemented

manual page unknown

> Scenario Length
>
> 6:00 am, Sept. 1 through 8:00 am, Sept. 3, 22 Game Turns.

File	Symbol	Code Snippet
omdurman-rules/src/turn_track.rs:11 GH:omdurman-	GameTime	`/// starts at one of these twelve times.` `#[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)]` `pub enum GameTime {`

rules/src/ turn_track.rs:11		`12 │     SixAM,` `13 │     EightAM,`
omdurman-rules/src/turn_track.rs:47 GH:omdurman-rules/src/turn_track.rs:47	**TurnEntry**	`45 │ /// A single entry on the Turn Record Track (rulebook §9.12, §9.22).` `46 │ #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)]` `47 │ pub struct TurnEntry {` `48 │     /// 1-based turn number.` `49 │     pub turn: u8,`
omdurman-rules/src/turn_track.rs:95 GH:omdurman-rules/src/turn_track.rs:95	**CAMPAIGN_TURN_TI**	`93 │ /// is turn 9, which carries the once-per-game Dervish Desertion Roll (§8.2) --` `94 │ /// the printed track prints "Dervish Desertion Roll / NIGHT" on that cell.` `95 │ pub const CAMPAIGN_TURN_TRACK: [TurnEntry; 22] = [` `96 │     // Row 1, left->right: Sept 1, 6 am -> 8 pm, then the first NIGHT.` `97 │     entry(1, SixAM, Day, TurnEvent::None),`
omdurman-rules/src/turn_track.rs:242 GH:omdurman-rules/src/turn_track.rs:242	**TurnLabel**	``240 │ /// printed cell; `Blank` is for unused positions in the 9×3 grid.`` `241 │ #[derive(Serialize, Deserialize, Debug, Clone, Copy, PartialEq, Eq)]` `242 │ pub enum TurnLabel {` `243 │     Blank,` `244 │     /// "SEPT. 1\n6:00 am" -- day header plus time (turn 1).`

Covered by tests: **omdurman-rules::src::turn_track::campaign_track_22_turns** **omdurman-rules::src::turn_track::desertion_on_first_night** **omdurman-rules::src::turn_track::campaign_track_label_and_day_night_agree** **omdurman-rules::src::turn_track::game_time_display_all_variants** **omdurman-rules::src::turn_track::turn_label_display** **omdurman-rules::src::turn_track::turn_label_out_of_range_is_none** **omdurman-rules::src::effects::tests::game_over_after_campaign_turns**

§9.13 – Special Rules (Campaign)

descriptive

manual page unknown

> Special Rules
>
> None.

§9.14 – Victory Conditions (Campaign)

implemented

manual page unknown

> Victory Conditions
>
> The Mahdi's Tomb in Omdurman was not only the tallest structure in the entire Sudan in 1898, it was also a Dervish holy shrine. Its loss or destruction would be a severe blow to the Mahdist cause. It is accordingly assigned 25 victory points which are awarded to the player who controls it at the conclusion of play. The Dervish player controls it at the start of play. As a tactical note, the Anglo-Egyptian player will find a decisive victory almost impossible unless he takes the Mahdi's Tomb from the Dervish player. To take the Tomb hex, it must be occupied by one British leader plus any one non-"Friendlies" Anglo-Egyptian combat unit (both undisrupted) at the conclusion of play.
>
> Additional victory points are awarded as follows:
>
> **Dervish Player receives:**
> - 10 pts: each British leader eliminated

(see manual for full text)

See also: §5.21

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:1168 GH:omdurman-rules/src/lib.rs:1168	**VpSource**	1166 | /// the manual and the engine. 1167 | #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 1168 | pub enum VpSource { 1169 | // ----- Anglo-Egyptian player receives: 1170 | /// Mahdi's Tomb control at conclusion of play (§9.14).
omdurman-rules/src/lib.rs:1194 GH:omdurman-rules/src/lib.rs:1194	**VpSource::points**	1192 | impl VpSource { 1193 | /// VP awarded to `who_scores()` (rulebook §9.14). 1194 | pub fn points(self) -> VictoryPoints { 1195 | match self { 1196 | VpSource::MahdisTomb => VictoryPoints::new(25),
omdurman-rules/src/lib.rs:1209 GH:omdurman-rules/src/lib.rs:1209	**VpSource::who_scores**	1207 | 1208 | /// Which player receives these victory points (rulebook §9.14). 1209 | pub fn who_scores(self) -> Player { 1210 | match self { 1211 | VpSource::MahdisTomb
omdurman-rules/src/lib.rs:1244 GH:omdurman-rules/src/lib.rs:1244	**VictoryLedger**	1242 | /// Cumulative victory ledger for one scenario (rulebook §9.14). 1243 | #[derive(Serialize, Deserialize, Clone, Debug, Default)] 1244 | pub struct VictoryLedger { 1245 | pub events: Vec<VpEvent>, 1246 | }
omdurman-rules/src/lib.rs:1250 GH:omdurman-rules/src/lib.rs:1250	**VpEvent**	1248 | /// A single victory-point scoring event (rulebook §9.14). 1249 | #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 1250 | pub struct VpEvent { 1251 | pub turn: GameTurnIndex, 1252 | pub source: VpSource,
omdurman-rules/src/lib.rs:1257 GH:omdurman-rules/src/lib.rs:1257	**VictoryLedger::total_for**	1255 | impl VictoryLedger { 1256 | /// Total victory points earned by a given player (rulebook §9.14). 1257 | pub fn total_for(&self, player: Player) -> VictoryPoints { 1258 | VictoryPoints(1259 | self.events
omdurman-rules/src/lib.rs:1269 GH:omdurman-rules/src/lib.rs:1269	**VictoryLedger::superiority**	1267 | /// Net superiority: positive = Anglo-Egyptian ahead, negative = Dervish ahead 1268 | /// (rulebook §9.14). 1269 | pub fn superiority(&self) -> VictoryPoints { 1270 | VictoryPoints(1271 | self.total_for(Player::AngloEgyptian).value() - self.total_for(Player::Dervish).value(),
omdurman-rules/src/lib.rs:1289 GH:omdurman-rules/src/lib.rs:1289	**CampaignVictoryLevel**	1287 | /// Campaign-game victory levels (§9.14). 1288 | #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 1289 | pub enum CampaignVictoryLevel { 1290 | Draw, 1291 | Marginal(Player),
omdurman-rules/src/lib.rs:1298 GH:omdurman-rules/src/lib.rs:1298	**CampaignVictoryLevel::f**	1296 | impl CampaignVictoryLevel { 1297 | /// Assign a level from the net superiority (§9.14). 1298 | pub fn from_superiority(s: VictoryPoints) -> Self { 1299 | let net = s.0; 1300 | // Positive -> Anglo-Egyptian thresholds: 15/30/50

<table>
<tr><td>omdurman-rules/src/effects/movement.rs:194
GH:omdurman-rules/src/effects/movement.rs:194</td><td>score_elimination</td><td><pre>192 | .find_unit(leader)
193 | .is_some_and(|u|
u.profile.identity.is_gordon());
194 | score_elimination(state, leader,
ElimCause::Combat);
195 | state.units.retain(|u| u.id != leader);
196 | if is_gordon {</pre></td></tr>
<tr><td>omdurman-rules/src/lib.rs:239
GH:omdurman-rules/src/lib.rs:239</td><td>VictoryPoints</td><td><pre>237 | /// (rulebook §9.14).
238 | #[derive(Serialize, Deserialize, Clone, Copy,
PartialEq, Eq, PartialOrd, Ord, Debug, Default)]
239 | pub struct VictoryPoints(i32);
240 |
241 | impl VictoryPoints {</pre></td></tr>
</table>

Proven by: **omdurman-rules::src::lib::campaign_victory_levels_match_manual_superiority_table**

Covered by tests: **omdurman-rules::src::effects::tests::friendlies_bank_scores_by_side** **omdurman-rules::src::effects::tests::mahdis_tomb_not_scored_without_a_leader** **omdurman-rules::src::effects::tests::mahdis_tomb_scores_for_anglo_egyptian_when_held** **omdurman-rules::src::lib::vp_source_attributes**

§9.21 – Set Up (Historical)

descriptive

manual page unknown

> Set Up

§9.22 – Scenario Length (Historical)

implemented

manual page unknown

> Scenario Length
>
> 6:00 am, September 2 through 12:00 noon, September 2. Four game turns.

<table>
<tr><th>File</th><th>Symbol</th><th>Code Snippet</th></tr>
<tr><td>omdurman-rules/src/turn_track.rs:132
GH:omdurman-rules/src/turn_track.rs:132</td><td>HISTORICAL_TURN_</td><td><pre>130 |
131 | /// Historical scenario track (§9.22 -- 4 turns, Sept 2
6:00 am -> 12:00 pm).
132 | pub const HISTORICAL_TURN_TRACK: [TurnEntry; 4] = [
133 | TurnEntry {
134 | turn: 1,</pre></td></tr>
</table>

Covered by tests: **omdurman-rules::src::turn_track::historical_turn_all_four_turns**

§9.23 – Special Rule: The Zariba

descriptive

manual page unknown

> Special Rule: “The Zariba”
>
> **9.231) Thorn hedge hexsides:** –2 to die roll on all Dervish fire attacks; may not melee across in either direction; may not advance after combat across in either direction.
>
> **9.232) Trench hexsides:** –4 to die roll on all Dervish fire attacks vs. entrenched units only; –2 (instead of +2) melee modifier to Dervish units melee attacking an entrenched unit; entrenched units may be fired “over” in both directions (i.e. they do not block line of sight); units are considered “entrenched” if they are directly adjacent to (and on the Nile River side of) a trench hexside.

>

(see manual for full text)

See also: §9.231, §9.232

§9.24 – Victory Conditions (Historical)

implemented

manual page unknown

> Victory Conditions
>
> Victory Levels are based solely on eliminating enemy units while conserving your own force as much as possible.
>
> | Victory Level | Anglo-Egyptian Player (Dervish units eliminated) | Dervish Player (Anglo-Egyptian units eliminated) | |—|—|—| | 5 — DECISIVE | 100+ | 30+ | | 4 — STRATEGIC | 60–99 | 15–29 | | 3 — TACTICAL | 45–59 | 10–14 | | 2 — MARGINAL | 30–44 | 5–9 | | 1 — DRAW | 0–29 | 0–4 |
>
> The lower value victory level is then subtracted from the higher level to determine a player's net victory. For example, if the Anglo-Egyptian player eliminates 104 Dervish units (decisive victory) but loses 18 units doing it (Dervish Strategic), the Anglo-Egyptian player only nets out with a draw (decisive worth 5 minus strategic worth 4 = 1, draw).

(see manual for full text)

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:1328 GH:omdurman-rules/src/lib.rs:1328	HistoricalVictoryLevel	`1326 | /// draw").` `1327 | #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Debug)]` `1328 | pub enum HistoricalVictoryLevel {` `1329 |     Draw = 1,` `1330 |     Marginal = 2,`

Proven by: **omdurman-rules::src::lib::historical_victory_ladders_match_manual_bands**

Covered by tests: **omdurman-rules::src::lib::historical_victory_level_for_dervish** **omdurman-rules::src::lib::historical_victory_level_for_anglo_egyptian**

§9.31 – Bonus game map

implemented

manual page unknown

> Only the small FALL OF KHARTOUM scenario map is used for this game.

File	Symbol	Code Snippet
omdurman-rules/src/board_data.rs:37 GH:omdurman-rules/src/board_data.rs:37	fall_of_khartoum_map_	`35 |` `36 | /// The Fall-of-Khartoum board (§9.3).` `37 | pub fn fall_of_khartoum_map_data() -> MapData {` `38 |     FOK.get_or_init(|| parse(FOK_RON)).clone()` `39 | }`

Covered by tests: **omdurman-app::src::tests::scenario_maps_to_board** **omdurman-app::src::tests::start_game_scenario_selects_board**

§9.32 – Set Up (Bonus)

descriptive

manual page unknown

> Set Up

§9.33 – Scenario Length (Bonus)

implemented

manual page unknown

> Scenario Length
>
> Variable, see victory conditions (9.35). Rarely lasts five turns.

See also: §9.35

File	Symbol	Code Snippet
omdurman-rules/src/turn_track.rs:172 GH:omdurman-rules/src/turn_track.rs:172	**FALL_OF_KHARTOUM**	`/// (the rulebook fixes none); only `day_night` is rule-bearing (night halves` `/// Anglo-Egyptian movement and ranges and bars howitzer fire, §8.1).` `pub const FALL_OF_KHARTOUM_TURN_TRACK: [TurnEntry; 8] = [` `    TurnEntry {` `        turn: 1,`

Covered by tests: **omdurman-rules::src::turn_track::fall_of_khartoum_turn_one_is_night** **omdurman-rules::src::turn_track::fall_of_khartoum_turns_3_to_8_are_day**

§9.34 – Special Rules (Bonus)

descriptive

manual page unknown

> Special Rules

§9.35 – Victory Conditions (Bonus)

implemented

manual page unknown

> Victory Conditions
>
> Victory is determined by how many turns it takes the Dervish player to eliminate the GORDON leader unit and how many Dervish units are eliminated:
>
> - Dervish decisive: eliminate GORDON turn four or sooner.
> - Dervish tactical: eliminate GORDON turn five.
> - Dervish marginal: eliminate GORDON turn six.
> - British marginal: GORDON survives end of turn six.
> - British tactical: GORDON survives end of turn seven.
> - British decisive: GORDON survives end of turn eight.

(see manual for full text)

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:1446 GH:omdurman-rules/src/lib.rs:1446	**FoKVictoryLevel::resolve**	`    /// rulebook: GORDON dies turn 5 (tactical) with 24 Dervish losses (−2` `    /// levels) nets a British marginal.` `    pub fn resolve(gordon_died_turn: Option<u8>, scenario_end_turn: u8, dervish_lost: i16) -> Self {` `        let base = Self::base(gordon_died_turn,`

		``scenario_end_turn);``` ```1448 │ let base_idx = Self::LADDER```

Proven by: **omdurman-rules::src::lib::fok_victory_ladder_penalties_shift_monotonically**

Covered by tests: **omdurman-rules::src::lib::fok_victory_level_worked_example** **omdurman-rules::src::lib::fok_victory_level_gordon_died_early** **omdurman-rules::src::lib::fok_victory_level_late_gordon_death** **omdurman-rules::src::lib::fok_victory_level_gordon_survived**

§9.111 – Dervish set up (Campaign)

implemented

manual page unknown

> Dervish player sets up first, moves second.
>
> - Isa Zachneih infantry unit: anywhere on the east bank, in or south of El Debeba.
> - KHALIFA ABDULLAH: in the walled city of Omdurman, in either palace hex.
> - 3 artillery units, and all Taiasha units: anywhere in the walled city of Omdurman.
> - 17 forts: anywhere on the mapsheet south of the Khor Shambat on the west bank, and/or south of all Halfaya hut hexes on the east bank and Nile River islands.
> - 2 gunboats: any south edge Nile River hexes.
>
> **9.112) Dervish reinforcements:** all reinforcements enter on the west edge of the mapsheet, south of the Khor Shambat. Each unit pays the terrain cost of the hex through which it enters, no matter how many units enter through

(see manual for full text)

See also: §9.112

File	Symbol	Code Snippet
omdurman-rules/src/effects/state.rs:243 GH:omdurman-rules/src/effects/state.rs:243	**setup_complete**	```241 │ /// currently shares the same "both sides deployed" gate; when a scenario``` ```242 │ /// needs a different minimum, branch on `self.scenario` here.``` ```243 │ pub fn setup_complete(&self) -> Result<(), RuleError> {``` ```244 │ let has = |player| {``` ```245 │ self.units```

Covered by tests: **omdurman-app::src::scenario_setup::campaign_has_no_fixed_placements** **omdurman-rules::src::unit_profiles::hadendowa_first_cell_is_isa_zachneih** **omdurman-rules::src::effects::tests::campaign_deployment_is_boat_land_exclusive** **omdurman-rules::src::effects::tests::campaign_setup_rejects_non_initial_force**

§9.112 – Dervish reinforcements (Campaign)

implemented

manual page unknown

File	Symbol	Code Snippet
omdurman-rules/src/effects/effect.rs:182 GH:omdurman-rules/src/effects/effect.rs:182	**PlaceReinforcements**	```180 │ // -- Reinforcement / placement --``` ```181 │ /// Place reinforcements onto the map (rulebook §9.112, §9.113).``` ```182 │ PlaceReinforcements(Vec<UnitPlacement>),``` ```183 │``` ```184 │ // -- Scenario-specific ---```
omdurman-rules/src/effects/	**apply_place_reinforceme**	```42 │ GameEffect::Demolition { unit_id, target } => apply_demolition(state, *unit_id, *target),```

dispatch.rs:44 GH:omdurman-rules/src/effects/dispatch.rs:44		`43 │         GameEffect::PlaceReinforcements(placements) => {` `44 │             apply_place_reinforcements(state, placements)` `45 │         }` `46 │         GameEffect::DervishDesertion { roll, deserters } => {`
omdurman-rules/src/reinforcements.rs:69 GH:omdurman-rules/src/reinforcements.rs:69	**dervish_campaign_sched**	`67 │ /// All reinforcements enter on the west edge, south of the Khor Shambat.` `68 │ /// Each unit pays terrain cost of the hex it enters through.` `69 │ pub fn dervish_campaign_schedule() -> ReinforcementSchedule {` `70 │     ReinforcementSchedule {` `71 │         player: Player::Dervish,`
omdurman-types/src/lib.rs:524 GH:omdurman-types/src/lib.rs:524	**Location**	`522 │ /// Named map landmarks (rulebook mapsheet, §9.111, §9.113, §9.212 scenarios).` `523 │ #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug, strum::Display)]` `524 │ pub enum Location {` `525 │     FortMakran,` `526 │     NorthFort,`
omdurman-types/src/lib.rs:604 GH:omdurman-types/src/lib.rs:604	**SetupLetter**	`602 │ /// Each letter marks a specific hex where a Dervish leader is placed.` `603 │ #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug, strum::Display)]` `604 │ pub enum SetupLetter {` `605 │     Y,` `606 │     K,`
omdurman-types/src/lib.rs:736 GH:omdurman-types/src/lib.rs:736	**Faction**	``734 │ /// `Some(BrigadeId::friendlies())`.`` `735 │ #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)]` `736 │ pub enum Faction {` `737 │     Dervish {` `738 │         tribe: DervishTribe,`

Covered by tests: **omdurman-rules::src::reinforcements::dervish_schedule_has_three_waves** **omdurman-rules::src::reinforcements::dervish_wave_one_has_baggaara_and_three_leaders** **omdurman-rules::src::reinforcements::dervish_wave_two_has_hadendowa** **omdurman-rules::src::reinforcements::dervish_wave_three_is_all_remaining** **omdurman-rules::src::reinforcements::wave_for_turn_returns_correct_wave** **omdurman-rules::src::effects::tests::campaign_reinforcements_gate_by_wave** **omdurman-rules::src::effects::tests::campaign_setup_rejects_non_initial_force**

§9.113 – Anglo-Egyptian reinforcements (Campaign)

implemented

manual page unknown

> The Anglo-Egyptian player moves first. There are no Anglo-Egyptian units on the mapsheet at start. The GORDON unit is not used in this scenario.
>
> - The leader units KITCHENER, GATACRE, and HUNTER may enter anytime during the first four game turns and do not count against the 12 units per turn limit. All three leaders must be in play by the end of turn four!
> - All gunboats enter through any north edge Nile River hex, paying one movement point for the first hex entered. The “Friendlies” brigade enters through the Abu Alim hut hex on the east bank, paying eight movement points per unit. All other Anglo-Egyptian units enter through the west bank “ANGLO-EGYPTIAN ENTRANCE AREA”, each unit paying one movement point to enter the mapsheet.
> - Turn 1) Any three gunboats; “Friendlies” brigade; Egyptian Cavalry; Horse Artillery; and two infantry brigades

(see manual for full text)

File	Symbol	Code Snippet
omdurman-rules/src/reinforcements.rs:126 GH:omdurman-rules/src/reinforcements.rs:126	**anglo_egyptian_campaig**	`124 | /// - "Friendlies" enter via Abu Alim hut on the east bank (8 MP per unit).` `125 | /// - All other AE units enter via the Anglo-Egyptian Entrance Area (1 MP).` `126 | pub fn anglo_egyptian_campaign_schedule() -> ReinforcementSchedule {` `127 |     let free_leaders = vec![` `128 |         CampaignLeader::British(BritishLeader::Kitchener),`

Covered by tests: **omdurman-rules::src::reinforcements::anglo_egyptian_schedule_has_four_waves** **omdurman-rules::src::reinforcements::anglo_egyptian_leaders_available_each_wave** **omdurman-rules::src::reinforcements::anglo_egyptian_turn_four_is_all_remaining** **omdurman-rules::src::effects::tests::campaign_reinforcement_cap_and_double_entry** **omdurman-rules::src::effects::tests::campaign_gunboats_quota_three_per_turn** **omdurman-rules::src::effects::tests::reinforcement_rejected_onto_enemy_occupied_hex** **omdurman-rules::src::effects::tests::campaign_setup_rejects_non_initial_force**

§9.211 – Anglo-Egyptian set up first, moves second (Historical)

implemented

manual page unknown

> The Anglo-Egyptian player sets up first, and moves second:
>
> - Not in play: GORDON leader unit, “Friendlies” brigade.
> - Gunboats start in any Nile River hexes adjacent to the Zariba.
> - Camel Corps, Egyptian Cavalry, and Horse Artillery start in the village of Kerreri hut hexes.
> - All remaining Anglo-Egyptian units set up in the 13 hexes of the Zariba.

File	Symbol	Code Snippet
omdurman-rules/src/effects/dispatch.rs:346 GH:omdurman-rules/src/effects/dispatch.rs:346	**first_player**	`344 |` `345 | /// The player who moves first in a scenario (§4, §9.113, §9.212, §9.322).` `346 | pub fn first_player(scenario: Scenario) -> Player {` `347 |     match scenario {` `348 |         Scenario::Campaign => Player::AngloEgyptian,`

Covered by tests: **omdurman-rules::src::effects::tests::historical_setup_rejects_not_in_play_units**

§9.212 – Dervish set up (Historical) – deployment zones and leader-to-hex mapping

implemented

manual page unknown

> The Dervish player sets up second, and moves first.
>
> - Not in play: Isa Zachneih, gunboats, and forts.
> - All Dervish units must be set up out of the line of sight of all Anglo-Egyptian units.
> - Dervish leaders start on the lettered hexes:
>   - A: Ali Wad Helu
>   - D: Sheik El Din
>   - Y: Yakub
>   - K: Khalifa Abdullah
>   - S: Sherif
>   - O: Osman Digna

(see manual for full text)

File	Symbol	Code Snippet
omdurman-rules/src/effects/state.rs:353 GH:omdurman-rules/src/effects/state.rs:353	**in_deployment_zone**	351 /// plan / UI rather than this hex predicate. Documented, not silently 352 /// dropped. 353 pub fn in_deployment_zone(&self, player: Player, hex: HexCoord, is_boat: bool) -> bool { 354 // No board attached -> permissive (unit tests, unbound session). 355 if self.board.terrain.is_empty() {
omdurman-rules/src/lib.rs:387 GH:omdurman-rules/src/lib.rs:387	**Dervish-Leader::setup_letter**	385 /// (§9.212): A→Ali Wad Helu, D→Sheik El Din, Y→Yakub, K→Khalifa Abdullah, 386 /// S→Sherif, O→Osman Digna. Inverse of [`dervish_leader_for_setup_letter`]. 387 pub fn setup_letter(self) -> SetupLetter { 388 match self { 389 DervishLeader::AliWadHelu => SetupLetter::A,
omdurman-rules/src/lib.rs:403 GH:omdurman-rules/src/lib.rs:403	**dervish_leader_for_setu**	401 /// inherent impl here, so the mapping is a free function -- the bijective 402 /// inverse of [`DervishLeader::setup_letter`]. 403 pub fn dervish_leader_for_setup_letter(letter: SetupLetter) -> DervishLeader { 404 match letter { 405 SetupLetter::A => DervishLeader::AliWadHelu,

Covered by tests: **omdurman-rules::src::effects::tests::deploy_rejected_outside_zone** **omdurman-rules::src::unit_profiles::embedded_leaders_resolve_from_their_host_section** **omdurman-app::src::scenario_setup::historical_places_all_six_leaders_when_anchors_present** **omdurman-app::src::scenario_setup::missing_anchor_is_reported_not_dropped_silently** **omdurman-rules::src::effects::tests::setup_letter_dervish_leader_roundtrip** **omdurman-rules::src::effects::tests::setup_letter_to_dervish_leader_known_values** **omdurman-rules::src::effects::tests::historical_setup_rejects_not_in_play_units**

§9.231 – Thorn hedge hexsides

implemented

manual page unknown

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:939 GH:omdurman-rules/src/lib.rs:939	**ZaribaThornHedge**	937 Terrain(i16), 938 /// -2 thorn-hedge defensive modifier (§9.231). 939 ZaribaThornHedge, 940 /// -4 trench defensive modifier (§9.232). Only applies vs. "entrenched" 941 /// units (those Nile-side of the trench hexside).
omdurman-types/src/lib.rs:206 GH:omdurman-types/src/lib.rs:206	**ZaribaThornHedge**	204 Crest, 205 /// Historical-scenario thorn-hedge segment of the Zariba (§9.231). 206 ZaribaThornHedge, 207 /// Historical-scenario trench segment of the Zariba (§9.232). 208 ZaribaTrench,
omdurman-rules/src/board.rs:284 GH:omdurman-rules/src/board.rs:284	**has_zariba_thorn_hedge**	282 /// hexside on its perimeter — i.e. whether the ZaribaThornHedge modifier 283 /// applies (§9.231). 284 pub fn has_zariba_thorn_hedge(&self, hex: HexCoord) -> bool { 285 for n in hex.neighbors() { 286 if let Some(kind) = self.hexside_between(hex, n)

Covered by tests: **omdurman-rules::src::effects::tests::zariba_fire_penalties_apply_to_dervish_fire_only**

§9.232 – Trench hexsides

implemented

manual page unknown

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:942 GH:omdurman-rules/src/lib.rs:942	**ZaribaTrenchEntrenched**	`940 │     /// -4 trench defensive modifier (§9.232). Only applies vs. "entrenched"` `941 │     /// units (those Nile-side of the trench hexside).` `942 │     ZaribaTrenchEntrenched,` `943 │ }` `944 │`
omdurman-types/src/lib.rs:208 GH:omdurman-types/src/lib.rs:208	**ZaribaTrench**	`206 │     ZaribaThornHedge,` `207 │     /// Historical-scenario trench segment of the Zariba (§9.232).` `208 │     ZaribaTrench,` `209 │     /// One of the two end hexsides of a Zariba trench segment that connect to` `210 │     /// the Nile River (§9.233).  Units may only enter/leave the Zariba via`

Proven by: **omdurman-rules::src::lib::melee_modifier_keeps_roll_legal**

Covered by tests: **omdurman-rules::src::effects::tests::trench_entrenched_units_take_trench_modifiers**

§9.233 – Zariba entry/exit costs

implemented

manual page unknown

> Units may only enter and/or leave the Zariba via the two end hexsides that connect to the Nile River, paying +2 movement points to cross (Exception: advance after combat across an entrenched hexside).

File	Symbol	Code Snippet
omdurman-types/src/lib.rs:258 GH:omdurman-types/src/lib.rs:258	**blocks_movement**	``256 │     /// `omdurman-rules`). The trench *end* variants are therefore intentionally`` `257 │     /// not blocking.` `258 │     pub fn blocks_movement(self) -> bool {` `259 │         matches!(` `260 │             self,`
omdurman-rules/src/board.rs:299 GH:omdurman-rules/src/board.rs:299	**zariba_entry_surcharge**	``297 │     /// movement points to cross"). Returns 2 when the edge between `from` and`` ``298 │     /// `to` is one of the two trench ends, else 0.`` `299 │     pub fn zariba_entry_surcharge(&self, from: HexCoord, to: HexCoord) -> i16 {` `300 │         match self.hexside_between(from, to) {` `301 │             Some(k) if k.is_zariba_trench_end() => 2,`
omdurman-rules/src/effects/state.rs:848 GH:omdurman-rules/src/effects/state.rs:848	**movement_cost_for**	`846 │     ///` `847 │     /// §5.42: entering or leaving an enemy ZOC adds no MP cost.` `848 │     pub fn movement_cost_for(` `849 │         &self,` `850 │         unit: &UnitPlacement,`
omdurman-rules/src/effects/movement.rs:31 GH:omdurman-rules/src/effects/movement.rs:31	**apply_move_unit**	`29 │ /// the true terrain cost (§5.11) and enforces gunboat upstream/downstream` ``30 │ /// allowances (§5.24); otherwise it falls back to the caller-supplied `cost`.`` `31 │ pub fn apply_move_unit(` `32 │     state: &mut GameState,` `33 │     unit_id: UnitId,`

Covered by tests: **omdurman-rules::src::effects::tests::zariba_end_hexside_costs_extra_mp** **omdurman-rules::src::effects::tests::zariba_thorn_hedge_blocks_movement**

§9.321 – British set up (Bonus)

implemented

manual page unknown

> The British player sets up first, moves second:
>
> - General GORDON leader unit in the palace.
> - Two old style (unnamed) gunboats in any Nile River hexes.
> - Set up in any building or hut hexes of Khartoum, Forts Makran and/or Buri, and/or adjacent to any wall hex:
>   - one Egyptian Battalion artillery unit
>   - two British infantry units (represents Caucasian troops)
>   - three Egyptian infantry units (represents Cairo "Bazouks")
>   - four Sudan infantry units (represents Sudanese blacks)
>   - four "Friendlies" units (represents the Shaggyeh)

(see manual for full text)

File	Symbol	Code Snippet
omdurman-rules/src/scenario_setup.rs:42 GH:omdurman-rules/src/scenario_setup.rs:42	**FALL_OF_KHARTOUM**	`40 │ /// North Fort uses a campaign HadendowaForts counter (one of the spare fort` `41 │ /// sprites).` `42 │ pub const FALL_OF_KHARTOUM_SETUP: &[FixedPlacement] = &[` `43 │     FixedPlacement {` `44 │         section: SectionName::BritishBoats,`

Covered by tests: **omdurman-rules::src::effects::tests::confirm_ready_rejected_below_scenario_target** **omdurman-rules::src::effects::tests::remove_deployed_unit_happy_path** **omdurman-app::src::scenario_setup::fall_of_khartoum_places_gordon_in_the_palace** **omdurman-app::src::scenario_setup::fall_of_khartoum_reports_missing_palace** **omdurman-app::src::scenario_setup::fall_of_khartoum_fort_landmarks_sit_at_the_correct_hexes** **omdurman-rules::src::effects::tests::fok_ae_gunboat_deploys_only_on_nile** **omdurman-rules::src::effects::tests::fok_ae_land_unit_rejected_on_nile** **omdurman-rules::src::effects::tests::fok_setup_complete_requires_full_oob** **omdurman-rules::src::effects::tests::deploy_via_real_sprite_resolution_matches_engine** **omdurman-rules::src::effects::tests::british_boats_named_vs_old_gunboat_detection** **omdurman-rules::src::effects::tests::fok_order_of_battle_british** **omdurman-rules::src::effects::tests::fok_order_of_battle_dervish** **omdurman-rules::src::effects::tests::fok_caps_bind_across_counter_variants**

§9.322 – Dervish enters turn one (Bonus)

implemented

manual page unknown

> Dervish player moves first: enters turn one through any hexes on the south or east edge of the map.
>
> - 32 Mulazmin units (represents combined forces of Wad El Nejumi, Abu Girgeh, and Sheik El Obeid)
> - 2 Hadendowa; 6 Kehena; 5 Degheim (represents Mahdi's combined west bank forces)
> - 3 Dervish artillery units.

File	Symbol	Code Snippet
omdurman-rules/src/unit_profiles.rs:340 GH:omdurman-	**ali_wad_helu**	`338 │ /// §9.2/§9.3) unplaceable -- the FoK order-of-battle table` `339 │ /// (`fok_cap_group`) is keyed by identity.` `340 │ pub fn ali_wad_helu(col: u32, row: u32) ->`

rules/src/ unit_profiles.rs:340		`Option<Classification> {` `    match (col, row) {` `        (0, 0) => dervish_leader(DervishLeader::AliWadHelu),`
omdurman-types/src/lib.rs:835 GH:omdurman-types/src/lib.rs:835	**sections_for_picker**	`    /// counter of those two forces and leave the §9.322 entry force -- and` `    /// with it the setup Ready gate (§9.2/§9.3) -- unplaceable.` `    pub fn sections_for_picker(self) -> Option<&'static [SectionName]> {` `        match self {` `            Scenario::Campaign | Scenario::Historical => None,`

Covered by tests: **omdurman-rules::src::unit_profiles::ali_wad_helu_block_resolves_leader_and_degelim_tribes** **omdurman-rules::tests::fok_setup_flow::degheim_counters_resolve_to_the_degheim_tribe** **omdurman-rules::src::effects::tests::fok_setup_complete_requires_full_oob** **omdurman-rules::src::effects::tests::fok_dervish_land_unit_rejected_on_nile** **omdurman-types::src::lib::fok_picker_allowlist_has_dervish_entry_force_blocks** **omdurman-rules::src::effects::tests::fok_order_of_battle_dervish** **omdurman-rules::src::effects::tests::fok_dervish_east_edge_on_diamond_board**

§9.341 – Turn 1 is always a night turn (Bonus)

implemented

manual page unknown

> Turn 1 is always a night turn (see 8.1).

See also: §8.1

File	Symbol	Code Snippet
omdurman-rules/src/turn_track.rs:172 GH:omdurman-rules/src/turn_track.rs:172	**FALL_OF_KHARTOUM**	`/// (the rulebook fixes none); only `day_night` is rule-bearing (night halves` `/// Anglo-Egyptian movement and ranges and bars howitzer fire, §8.1).` `pub const FALL_OF_KHARTOUM_TURN_TRACK: [TurnEntry; 8] = [` `    TurnEntry {` `        turn: 1,`

Covered by tests: **omdurman-rules::src::turn_track::fall_of_khartoum_turn_one_is_night**

§9.342 – All hexes are playable, including half hexes (Bonus)

implemented

manual page unknown

> All hexes are playable, including hexes showing up half or less.

File	Symbol	Code Snippet
omdurman-rules/src/board_data.rs:37 GH:omdurman-rules/src/board_data.rs:37	**fall_of_khartoum_map_**	`` `/// The Fall-of-Khartoum board (§9.3).` `pub fn fall_of_khartoum_map_data() -> MapData {` `    FOK.get_or_init(|| parse(FOK_RON)).clone()` `}`

Covered by tests: **omdurman-rules::src::effects::tests::fall_of_khartoum_board_excludes_no_hexes**

§9.343 – Both players use the Dervish Range Effects Table (Bonus)

implemented

manual page unknown

> Both players must use the Dervish Range Effects Table.

<table>
<tr><th>File</th><th>Symbol</th><th>Code Snippet</th></tr>
<tr><td>omdurman-rules/src/effects/fire.rs:53
GH:omdurman-rules/src/effects/fire.rs:53</td><td>range_band_for</td><td><pre>51 | /// in FALL OF KHARTOUM *both* players use the Dervish Range Effects Table
52 | /// (§9.343).
53 | pub fn range_band_for(
54 | scenario: Scenario,
55 | player: Player,</pre></td></tr>
</table>

Covered by tests: **omdurman-rules::src::effects::tests::fok_both_players_use_dervish_range_table**

§9.344 – Dervish controls the North Fort (Bonus)

implemented

manual page unknown

> The Dervish player controls the “North Fort” and may fire its guns.

<table>
<tr><th>File</th><th>Symbol</th><th>Code Snippet</th></tr>
<tr><td>omdurman-rules/src/scenario_setup.rs:42
GH:omdurman-rules/src/scenario_setup.rs:42</td><td>FALL_OF_KHARTOUM</td><td><pre>40 | /// North Fort uses a campaign HadendowaForts counter (one of the spare fort
41 | /// sprites).
42 | pub const FALL_OF_KHARTOUM_SETUP: &[FixedPlacement] = &[
43 | FixedPlacement {
44 | section: SectionName::BritishBoats,</pre></td></tr>
<tr><td>omdurman-rules/src/effects/state.rs:1387
GH:omdurman-rules/src/effects/state.rs:1387</td><td>hex_has_enemy_fort</td><td><pre>1385 | /// may neither occupy an enemy fort nor advance after combat into one
1386 | /// (forts are never captured -- only destroyed, §6.62/§6.53/§7.6).
1387 | pub fn hex_has_enemy_fort(&self, hex: HexCoord, mover: Player) -> bool {
1388 | self.units.iter().any(|u| {
1389 | u.position == hex</pre></td></tr>
</table>

Covered by tests: **omdurman-app::src::scenario_setup::fall_of_khartoum_places_gordon_in_the_palace** **omdurman-app::src::scenario_setup::fall_of_khartoum_fort_landmarks_sit_at_the_correct_hexes** **omdurman-app::src::scenario_setup::placement_done_gate_matches_by_identity_not_allocated_id** **omdurman-rules::src::effects::tests::fok_order_of_battle_dervish** **omdurman-types::src::lib::fok_picker_allowlist_has_dervish_entry_force_blocks**

§9.345 – Gunboat White Nile <-> Blue Nile crossing (Bonus)

implemented

manual page unknown

> The British gunboats may move from the White Nile to the Blue Nile and vice-versa at an off-board movement cost of six “upstream” movement points.

<table>
<tr><th>File</th><th>Symbol</th><th>Code Snippet</th></tr>
<tr><td>omdurman-rules/src/effects/state.rs:1371</td><td>is_nile_mouth_crossing</td><td><pre>1369 | /// must be named on the board, else this is `false` and the move falls
1370 | /// through to the ordinary contiguous-Nile rules.</pre></td></tr>
</table>

<table>
<tr><td>GH:omdurman-rules/src/effects/state.rs:1371</td><td></td><td><pre>1371 | pub fn is_nile_mouth_crossing(&self, from: HexCoord, to: HexCoord) -> bool {
1372 | let white = self
1373 | .board</pre></td></tr>
<tr><td>omdurman-types/src/lib.rs:540
GH:omdurman-types/src/lib.rs:540</td><td>Location::WhiteNileMouth</td><td><pre>538 | /// The off-board mouth of the White Nile branch (FALL OF KHARTOUM §9.345) --
539 | /// a British gunboat may cross to the Blue Nile mouth for 6 upstream MP.
540 | WhiteNileMouth,
541 | /// The off-board mouth of the Blue Nile branch (FALL OF KHARTOUM §9.345).
542 | BlueNileMouth,</pre></td></tr>
</table>

Covered by tests: **omdurman-rules::src::effects::tests::fok_gunboat_crosses_between_nile_mouths**

§9.346 – GORDON immobile, eliminated only at the Palace (Bonus)

implemented

manual page unknown

> The GORDON leader unit starts in the palace and may not move during the scenario. He may only be eliminated by a Dervish unit passing through or occupying the palace hex (as normal movement or as advance after combat).

<table>
<tr><th>File</th><th>Symbol</th><th>Code Snippet</th></tr>
<tr><td>omdurman-rules/src/effects/dispatch.rs:482
GH:omdurman-rules/src/effects/dispatch.rs:482</td><td>check_gordon_palace</td><td><pre>480 | /// after combat). Records the turn (which fixes the §9.35 victory level) and
481 | /// ends the game. A no-op outside FoK, or once GORDON is already gone.
482 | pub fn check_gordon_palace(state: &mut GameState) {
483 | if state.scenario != Scenario::FallOfKhartoum || state.gordon_eliminated_turn.is_some() {
484 | return;</pre></td></tr>
<tr><td>omdurman-rules/src/lib.rs:644
GH:omdurman-rules/src/lib.rs:644</td><td>UnitIdentity::is_gordon</td><td><pre>642 | /// Whether this is the GORDON leader unit (§9.32, §9.346) -- the immobile
643 | /// palace defender whose elimination ends FALL OF KHARTOUM (§9.35).
644 | pub fn is_gordon(&self) -> bool {
645 | matches!(
646 | self,</pre></td></tr>
</table>

Covered by tests: **omdurman-rules::src::unit_profiles::gordon_is_an_immobile_british_leader** **omdurman-rules::src::effects::tests::gordon_survives_means_no_elimination** **omdurman-app::src::scenario_setup::fall_of_khartoum_places_gordon_in_the_palace**

`7/10 implemented`

§10 – Optional Rules

§10 – Optional Rules

descriptive

manual page unknown

> Optional Rules (Campaign game only)
>
> It is suggested that the most intriguing employment of the following two options is to permit the Dervish player to have either one or the other, but the Anglo-Egyptian player doesn't know which one until he stumbles onto it. Players are advised that the employment of both optionals in the same game is not recommended.

§10.1 – River Mines

descriptive

manual page unknown

> River Mines
>
> The Khalifa twice tried (unsuccessfully) to submerge a powerful mine in the Nile to sink or damage British gunboats. This option assumes that both attempts were successful.

§10.2 – River Chain

descriptive

manual page unknown

> River Chain
>
> The Khalifa also tried (also unsuccessfully) to string a heavy chain across the Nile to stop or slow down the British gunboats. This option assumes the chain was emplaced.

§10.11 – Secretly record mine hexes

implemented

manual page unknown

> Prior to the commencement of play the Dervish player secretly records two Nile River hexes to be mined (the mines may not both be placed in the same hex). These hexes must be south of the E–W hexrow in which the Khor Shambat empties into the Nile.

File	Symbol	Code Snippet
omdurman-rules/src/effects/setup.rs:330 GH:omdurman-rules/src/effects/setup.rs:330	**apply_place_mine**	`328 │ /// Lay a river mine during setup (§10.11). Validated by` `329 │ /// [`GameState::can_place_mine`].` `330 │ pub fn apply_place_mine(state: &mut GameState, hex: HexCoord) -> Result<(), RuleError> {` `331 │     state.can_place_mine(hex)?;` `332 │     state.mines.push(MinePlacement {`
omdurman-rules/src/effects/state.rs:174 GH:omdurman-rules/src/effects/state.rs:174	**GameState::mines**	`172 │             friendlies_transport: None,` `173 │             optional_rules: Vec::new(),` `174 │             mines: Vec::new(),` `175 │             chain: None,` `176 │             board: BoardInfo::default(),`
omdurman-rules/src/	**OptionalRule**	`314 │ /// two should be in play (rulebook §10).` `315 │ #[derive(Serialize, Deserialize, Clone, Copy,`

lib.rs:316 GH:omdurman-rules/src/lib.rs:316		`PartialEq, Eq, Debug)]` `316 │ pub enum OptionalRule {` `317 │     RiverMines,` `318 │     RiverChain,`

Covered by tests: **omdurman-rules::src::effects::tests::mine_and_chain_limits_enforced_in_setup** **omdurman-rules::src::effects::tests::mines_and_chain_require_their_optional_rule**

§10.12 – Mine resolution

implemented

manual page unknown

> When a British gunboat enters a mined hex, the Dervish player must order it to stop as it has struck a mine. The Dervish player then resolves the effect of the mine's blast by rolling the ten-sided die:
>
> - 1–4: No effect
> - 5–7: Gunboat damaged, lost use of its engines and must drift two hexes per turn (with the current) for the rest of the game. No effect on guns or Maxims unless they drift out of range.
> - 8–10: Gunboat sunk!

File	Symbol	Code Snippet
omdurman-rules/src/effects/effect.rs:200 GH:omdurman-rules/src/effects/effect.rs:200	**RiverMine**	`198 │     // -- Optional rules` `----------------------------------------------------` `199 │     /// River mine resolution (rulebook §10.12).` `200 │     RiverMine {` `201 │         gunboat_id: UnitId,` `202 │         hex: HexCoord,`
omdurman-rules/src/effects/river.rs:146 GH:omdurman-rules/src/effects/river.rs:146	**apply_river_mine**	`144 │` `145 │ /// Apply a river-mine resolution (rulebook §10.12).` `146 │ pub fn apply_river_mine(` `147 │     state: &mut GameState,` `148 │     gunboat_id: UnitId,`
omdurman-rules/src/lib.rs:1120 GH:omdurman-rules/src/lib.rs:1120	**MineResult**	`1118 │ /// British gunboat enters a mined hex.` `1119 │ #[derive(Serialize, Deserialize, Clone, Copy,` `PartialEq, Eq, Debug)]` `1120 │ pub enum MineResult {` `1121 │     /// Roll 1-4: no effect.` `1122 │     NoEffect,`

Covered by tests: **omdurman-rules::src::effects::tests::mine_fires_once_and_spares_dervish**

§10.13 – Mines consumed after both rolled for

implemented

manual page unknown

> After both mines have been rolled for, no more are available.

File	Symbol	Code Snippet
omdurman-rules/src/effects/river.rs:146 GH:omdurman-rules/src/effects/river.rs:146	**apply_river_mine**	`144 │` `145 │ /// Apply a river-mine resolution (rulebook §10.12).` `146 │ pub fn apply_river_mine(` `147 │     state: &mut GameState,` `148 │     gunboat_id: UnitId,`

Covered by tests: **omdurman-rules::src::effects::tests::mine_fires_once_and_spares_dervish**

§10.14 – Dervish gunboats pass safely

implemented

manual page unknown

> The Dervish player's gunboats may pass through the mined hexes with no ill effect (he knows where the mines are).

File	Symbol	Code Snippet
omdurman-rules/src/effects/river.rs:146 GH:omdurman-rules/src/effects/river.rs:146	**apply_river_mine**	`144 │` `145 │ /// Apply a river-mine resolution (rulebook §10.12).` `146 │ pub fn apply_river_mine(` `147 │     state: &mut GameState,` `148 │     gunboat_id: UnitId,`

Covered by tests: **omdurman-rules::src::effects::tests::mine_fires_once_and_spares_dervish**

§10.21 – Secretly record chain hexes

implemented

manual page unknown

> Prior to the commencement of play the Dervish player secretly records a line of river hexes (not exceeding four hexes long) across which the chain is strung. The hexes must be south of the E–W hexrow in which the Khor Shambat empties into the Nile.

File	Symbol	Code Snippet
omdurman-rules/src/effects/state.rs:618 GH:omdurman-rules/src/effects/state.rs:618	**can_place_chain**	`616 │     /// Read-only check of a river-chain placement in setup (§10.21): Setup phase` `617 │     /// and at most [`MAX_CHAIN_HEXES`] hexes.` `618 │     pub fn can_place_chain(&self, hexes: &[HexCoord]) -> Result<(), RuleError> {` `619 │         self.require_setup_phase()?;` `620 │         // Optional-rule gate: the chain exists only when the River Chain option`
omdurman-rules/src/effects/setup.rs:341 GH:omdurman-rules/src/effects/setup.rs:341	**apply_place_chain**	`339 │ /// Lay (or replace) the river chain during setup (§10.21). Validated by` `340 │ /// [`GameState::can_place_chain`].` `341 │ pub fn apply_place_chain(state: &mut GameState, hexes: &[HexCoord]) -> Result<(), RuleError> {` `342 │     state.can_place_chain(hexes)?;` `343 │     state.chain = Some(ChainPlacement {`
omdurman-rules/src/effects/state.rs:1838 GH:omdurman-rules/src/effects/state.rs:1838	**MAX_CHAIN_HEXES**	`1836 │` `1837 │ /// Maximum contiguous Nile hexes the river chain may span (§10.21).` `1838 │ pub const MAX_CHAIN_HEXES: usize = 4;` `1839 │` `1840 │ //` `-----------------------------------------------------------------------------------`

Covered by tests: **omdurman-rules::src::effects::tests::mine_and_chain_limits_enforced_in_setup** **omdurman-rules::src::effects::tests::mines_and_chain_require_their_optional_rule**

§10.22 – Gunboat stops on chained hex

implemented

manual page unknown

> When a British gunboat enters a “chained” river hex it must stop and may move no further that turn.

File	Symbol	Code Snippet
omdurman-rules/src/effects/state.rs:887 GH:omdurman-rules/src/effects/state.rs:887	can_move_gunboat	`885 │     /// upstream movement allowance is their maximum for that turn." Chained Nile` `886 │     /// hexes stop the gunboat (§10.22).` `887 │     pub fn can_move_gunboat(` `888 │         &self,` `889 │         unit_id: UnitId,`

Covered by tests: **omdurman-rules::src::effects::tests::chain_stops_gunboat_until_sunk**

§10.23 – Sinking the chain

implemented

manual page unknown

> No gunboats (British or Dervish) may cross the chain until it has been sunk by the British player. He may sink the chain by a) having an infantry or cavalry unit spend one complete turn on either riverbank adjacent to a “chained” river hex, or b) firing at the chain with artillery and achieving a 3 or more on the Combat Results Table.

File	Symbol	Code Snippet
omdurman-rules/src/effects/river.rs:190 GH:omdurman-rules/src/effects/river.rs:190	apply_sink_chain	`188 │ /// Sink the river chain (rulebook §10.23). Marks the placed chain cleared so it` `189 │ /// no longer stops gunboats (§10.22).` `190 │ pub fn apply_sink_chain(state: &mut GameState) -> Result<(), RuleError> {` `191 │     match state.chain.as_mut() {` `192 │         Some(chain) if !chain.sunk => {`

Covered by tests: **omdurman-rules::src::effects::tests::chain_stops_gunboat_until_sunk**

`0/1 implemented`

§11 – Historical Notes

§11 – Historical Notes

descriptive

manual page unknown

> Historical Notes
>
> In 1881 Mohammed Ahmed Ibn Al-Sayid Abdullah, the son of an obscure carpenter in the hinterlands of the Sudan, proclaimed himself the “Mahdi” — the Messiah of the Islamic faith. His timing was propitious indeed. Since the early 1820's a corrupt Egypt, with the Sultan of Turkey’s blessing, had incessantly raped the Sudan, taking ivory and flooding the slave markets with some half million captured Sudanese blacks. By 1880, nearly 40,000 Egyptian troops occupied outposts scattered throughout the Sudan, enforcing Egypt’s hold on this lucrative ivory and slave trade and squeezing the native population dry through vicious and corrupt tax officials. All was controlled from Khartoum via the office of Governor General of the Sudan. The title had been held by a succession of individuals, including General Charles Gordon, whose appointment was an attempt to reinstate some rudimentary justice in the Sudan after France and Britain assumed joint political control of a bankrupt Egypt.
>
> By 1881, however, Gordon’s term had expired and a new Governor General, again corrupt and incompetent,

(see manual for full text)

See also: §9.1, §9.2, §9.3

`0/1 implemented`

Credits

§Credits – Credits

descriptive

manual page unknown

1/1 implemented

Combat Results Table (shared reference)

§CRT – Combat Results Table (shared by §6.22 fire and §7.7 melee)

implemented

manual page unknown

<table>
<tr><th>File</th><th>Symbol</th><th>Code Snippet</th></tr>
<tr><td>omdurman-rules/src/combat_results_table.rs:10
GH:omdurman-rules/src/combat_results_table.rs:10</td><td>FireFactorRow</td><td><pre>
  8 | serde::Serialize, serde::Deserialize, Clone, Copy,
PartialEq, Eq, PartialOrd, Ord, Hash, Debug,
  9 |)]
 10 | pub enum FireFactorRow {
 11 | /// 1-5 factors
 12 | Row01to05,
</pre></td></tr>
<tr><td>omdurman-rules/src/combat_results_table.rs:46
GH:omdurman-rules/src/combat_results_table.rs:46</td><td>from_total</td><td><pre>
 44 |
 45 | /// Determine which row a given total fire factor
falls into (rulebook §6.22).
 46 | pub fn from_total(total: u16) -> Self {
 47 | match total {
 48 | 0..=5 => FireFactorRow::Row01to05,
</pre></td></tr>
<tr><td>omdurman-rules/src/combat_results_table.rs:88
GH:omdurman-rules/src/combat_results_table.rs:88</td><td>combat_results_table</td><td><pre>
 86 | /// D = `Disrupt` (1/2 of target units, round up)
 87 | /// 1...5 = `Eliminate(n)` (that many units removed)
 88 | pub fn combat_results_table(row: FireFactorRow, roll:
DieRoll) -> CombatResult {
 89 | crate::tables_data::CRT[row.index()][(roll.value()
- 1) as usize]
 90 | }
</pre></td></tr>
<tr><td>omdurman-rules/src/lib.rs:1008
GH:omdurman-rules/src/lib.rs:1008</td><td>CombatResult</td><td><pre>
1006 | /// * `--` -- no effect
1007 | #[derive(Serialize, Deserialize, Clone, Copy,
PartialEq, Eq, Debug)]
1008 | pub enum CombatResult {
1009 | NoEffect,
1010 | Disrupt,
</pre></td></tr>
</table>

Proven by: **omdurman-rules::src::lib::disrupt_half_is_rounded_up** **omdurman-rules::src::combat_results_table::crt_eliminate_count_stays_within_printed_bounds** **omdurman-rules::src::combat_results_table::crt_is_monotone_in_the_die_roll_for_every_row**

Covered by tests: **omdurman-rules::src::combat_results_table::ae_combat_results_table_lowest_is_no_effect** **omdurman-rules::src::combat_results_table::ae_combat_results_table_highest_is_eliminate_5** **omdurman-rules::src::combat_results_table::ae_combat_results_table_progresses_with_roll** **omdurman-rules::src::combat_results_table::ae_combat_results_table_progresses_with_factor** **omdurman-rules::src::combat_results_table::fire_factor_row_boundaries** **omdurman-rules::src::combat_results_table::fire_factor_row_remaining_boundaries** **omdurman-rules::src::combat_results_table::fire_factor_row_index_sequential** **omdurman-rules::src::combat_results_table::crt_all_rows_monotone_non_decreasing** **omdurman-rules::src::combat_results_table::crt_cross_row_monotone_for_each_roll** **omdurman-rules::src::combat_results_table::crt_lowest_row_is_worst_highest_row_is_best** **omdurman-rules::src::combat_results_table::crt_eliminate_never_exceeds_5** **omdurman-rules::src::combat_results_table::crt_every_cell_matches_the_table**

`0/1 implemented`

Reference – Charts and Tables

§Reference – Charts and Tables

out-of-scope

manual page unknown

Appendix: Symbol Index

Symbol	Sections
AdvanceAfterCombat	§6.82, §7.6
AdvancePhase	§4
AngloEgyptianDirectFire	§6.24
AngloEgyptianStandard	§7.7
ArtilleryBreachWall	§6.63
BattalionOrdinal	§5.54
Breach	§6.63
BrigadeId	§2.3, §5.54
BrigadeIntegrity	§5.54, §6.24
BritishLeader	§6.51
CAMPAIGN_TURN_TRACK	§9.12
CampaignVictoryLevel	§9.14
CombatResult	§CRT
ConstructZariba	§5.3
DayNight	§8.1
Demolition	§6.53
DemolitionTarget	§6.53
DervishDesertion	§8.2
DervishGunboat	§2.32
DervishLeaderCommandMis-match	§5.53
DervishStandard	§7.7
DervishTribe	§7.4
DervishTribeMix	§5.52
DervishVsTrenchedDefender	§7.7
DirectFire	§6.41
EnemyCohabitation	§5.51
FALL_OF_KHARTOUM_SETUP	§9.321, §9.344
FALL_OF_KHARTOUM_TURN_TRACK	§9.33, §9.341
Faction	§9.112
FireAttack	§6.54
FireCombat	§6.4
FireFactor	§6.11
FireFactorRow	§CRT
FireSubPhase	§6.4
Fort	§5.25, §6.54
Friendlies	§6.52
FriendliesAction	§5.21
FriendliesTransport	§5.21
GameState	§4
GameTime	§9.12

GameTurnIndex	§4
GunboatId	§2.32, §6.64
GunboatMovement	§5.24
GunboatStack	§5.51
HISTORICAL_TURN_TRACK	§9.22
HexDirection	§5.11
HexDistance	§6.22, §7.5
HexsideKind	§5.23
HexsideRef	§5.23
HistoricalVictoryLevel	§9.24
HowitzerFire	§6.64
Immobile	§5.25
Khor	§5.44
Location	§9.112
LosCondition	§6.3
LosFeature	§6.3
LosLevel	§6.3
MAX_CHAIN_HEXES	§10.21
MaximSecondAndHowitzer	§6.42
MeleeAttack	§7.7
MeleeCombat	§7.3
MeleeFactor	§7.1
MeleeModifier	§7.7
MineResult	§10.12
MovementAllowance	§5.11
MovementPoints	§5.11
NamedGunboat	§2.32
Old	§2.32
OldGunboat	§2.32
OptionalRule	§10.11
OverLimit	§5.51
PendingMelee	§4, §7
Phase	§4
PlaceReinforcements	§9.112
RangeBand	§6.16, §6.22
RetreatBeforeMelee	§7.5
RiverMine	§10.12
RoyalEngineers	§6.53
ScatterHexDirection	§6.64
SetupLetter	§9.112
SpriteAnnotation	§2.3
Terrain	§6.23
TransportState	§5.21
TurnEntry	§9.12

can_advance_after_combat	§6.7, §6.82, §7.6
can_fire_at	§6.64
can_fire_at_wall	§6.63
can_melee	§7.2, §7.4
can_move_gunboat	§10.22
can_move_unit	§5.12
can_move_unit_to	§5.22, §5.26, §5.43
can_place_chain	§10.21
can_retreat_before_melee	§7.5
check_gordon_palace	§9.346
check_stacking	§5.51
combat_results_table	§CRT
constructing_zariba	§5.3
defense_modifier	§6.23
demolishing	§6.53
demolition_targets	§6.53
dervish_campaign_schedule	§9.112
dervish_leader_for_setup_letter	§9.212
dervish_range_effects	§6.22
dervish_stacking_group	§5.52
dervish_tribe	§2.31
die_modifier	§6.24
effective_movement_at_night	§8.1
end_player_turn	§4, §5.13
fall_of_khartoum_map_data	§9.31, §9.342
fires_twice	§6.42
first_player	§9.211
friendlies_transport_offer	§5.21
from_superiority	§9.14
from_total	§CRT
halve	§8.1
has_combat_factors	§6.51
has_los	§6.21, §6.3
has_road	§5.11
has_zariba_thorn_hedge	§9.231
hex_has_enemy_fort	§9.344
hex_in_enemy_zoc	§5.26, §5.43, §5.44
howitzer_scatter	§6.64
in_deployment_zone	§5.22, §9.212
is_boat	§5.24
is_crossroad	§5.11
is_friendlies	§5.21, §6.52
is_gordon	§9.346
is_los_trees	§6.3